

ICPC Team Notebook

UIT.BTC

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1 Data Structure

1.1 Iterative Segment Tree

Complexity: $\mathcal{O}(\log n)$ per query/update, $\mathcal{O}(n)$ build
Fast non-recursive segment tree with point update, range query, and Dual SegTree (Range update + Point query)

```
1 // Iterative SegTree (0-indexed [l, r], Point update, Range max
  query, Tree Walk, Dual)
2 const int N = 2e5 + 5;
3 int n, sz, tree[4 * N];
4
5 void init(int _n) {
6     n = _n; sz = 1;
7     while (sz < n) sz <= 1;
8     fill(tree, tree + 2 * sz, 0);
9 }
10 void build(int a[]) {
11     init(n);
12     for (int i = 0; i < n; i++) tree[sz + i] = a[i];
13     for (int i = sz - 1; i > 0; i--) tree[i] = max(tree[i << 1],
14         tree[i << 1 | 1]);
15 }
16 void update(int p, int val) { // a[p] = val
17     for (tree[p += sz] = val; p > 1; p >= 1) tree[p >> 1] = max(
18         tree[p], tree[p ^ 1]);
19 }
20 int query(int l, int r) { // max [l, r]
21     int res = 0;
22     for (l += sz, r += sz + 1; l < r; l >= 1, r >= 1) {
23         if (l & 1) res = max(res, tree[l++]);
24         if (r & 1) res = max(res, tree[--r]);
25     }
26     return res;
27 }
28 // Tree Walk  $\mathcal{O}(\log N)$ : First index  $i \geq l$  with  $tree[i] \geq val$  (or n
  if none)
29 int max_right(int l, int val) {
30     int cur = 0;
31     for (l += sz; ; l >= 1) {
32         if (l & 1) {
33             if (max(cur, tree[l]) >= val) {
34                 while (l < sz) {
35                     l <= 1;
36                     if (max(cur, tree[l]) < val) cur = max(cur,
37                         tree[l++]);
38                 }
39                 return min(n, l - sz);
40             }
41             cur = max(cur, tree[l++]);
42         }
43         if ((l & (l - 1)) == 0) break;
44     }
45     return n;
46 }
47 // Tree Walk  $\mathcal{O}(\log N)$ : Last index  $i \leq r$  with  $tree[i] \geq val$  (or -1
  if none)
48 int min_left(int r, int val) {
49     int cur = 0;
50     for (r += sz + 1; ; r >= 1) {
51         if (r & 1) {
52             --r;
53             if (max(tree[r], cur) >= val) {
54                 while (r < sz) {
55                     r = r << 1 | 1;
56                     if (max(tree[r], cur) < val) cur = max(tree[r
57                         --], cur);
58                 }
59             }
60         }
61     }
62 }
```

```

55         return r - sz;
56     }
57     cur = max(tree[r], cur);
58 }
59 if ((r & (r + 1)) == 0) break;
60 }
61 return -1;
62 }
63 // Dual SegTree: Range update [l, r] += val, Point query at p
64 void update_range(int l, int r, int val) {
65     for (l += sz, r += sz + 1; l < r; l >= 1, r >= 1) {
66         if (l & 1) tree[l++] += val;
67         if (r & 1) tree[--r] += val;
68     }
69 }
70 int query_point(int p) {
71     int res = 0;
72     for (p += sz; p > 0; p >= 1) res += tree[p];
73     return res;
74 }

```

1.2 Lazy Segment Tree (Implementation Tips)

Lazy propagation discipline, tag compositions (Affine, Range Set), and $O(\log n)$ Tree Walk

- **Push/Pull Discipline:** Call `push(id, l, r)` at the start of `update / query`; call `pull(id)` (`st[id] = merge(st[2*id], st[2*id+1])`) after updating children.

- **Multiple Lazy Tags:**

- **Affine** ($x \mapsto a \cdot x + b$): Composing new (c, d) on $(a, b) \implies c(ax + b) + d = (ca)x + (cb + d)$.
Tag: `laz_a = laz_a * c; laz_b = laz_b * c + d;`
Node: `val = val * c + (r - l + 1) * d;`
- **Range Set vs Range Add:** Range Set ($x \leftarrow v$) overrides previous Range Add $\implies$ `reset add = 0.`

- **Tree Walk** ($O(\log n)$): If left child satisfies condition ($\max \geq X$ or $\text{sum} \geq S$), push lazy and recurse left; else adjust target and recurse right.

- **Bounds:** Allocate $4N$ for 1-based indexing (`id < 1` and `id < 1 | 1`).

1.3 Sparse Segment Tree

Complexity: $O(\log n)$ per query

Dynamic Segment Tree allocating nodes on demand

```

1 struct Node{
2     int l, r, ll, rr; // [l, r] are the ranges | ll, rr: left, right
        children
3 };
4 struct sparseIT{
5     /*A sample sparse IT with following operations:
6     - Get the number of elements in a range.
7     - Get the sum of elements in a range.
8     Lazy operations could be implemented as well
9     */
10    vector<Node> st;
11    int rt = 0;
12    int getInfo(int nn){
13        //Return Some kind of info
14    }
15    void update(int nn){
16        //Update the node nn.
17    }
18    sparseIT(int _l = 0, int _r = 0){

```

```

19    rt = 0;
20    st.pb(Node(_l, _r));
21 }
22 void updatePos(int nn, int pos, int value){
23     if (st[nn].l == st[nn].r){
24
25         //Change update position
26         return;
27     }
28     int mid = (st[nn].l + st[nn].r) >> 1;
29     if (pos <= mid){
30         if (st[nn].ll == -1) { //init
31             st[nn].ll = st.size();
32             st.pb(Node(st[nn].l, mid));
33         }
34         updatePos(st[nn].ll, pos, value);
35     }
36     else{
37         if (st[nn].rr == -1) { //init
38             st[nn].rr = st.size();
39             st.pb(Node(mid + 1, st[nn].r));
40         }
41         updatePos(st[nn].rr, pos, value);
42     }
43     update(nn);
44 }
45 int getRange(int nn, int u, int v){
46     if (u > v) return 0;
47     if (nn == -1) return 0;
48     if (st[nn].l >= u and st[nn].r <= v){
49         //return infos here
50     }
51     if (st[nn].l > v or st[nn].r < u) {
52         return 0;
53     }
54     return getSize(st[nn].ll, u, v) +
55         getSize(st[nn].rr, u, v);
56 }
57 };

```

1.4 Cartesian Tree

Complexity: $O(n)$ build via Monotonic Stack

Binary tree combining heap-order on values and BST-order on indices

- **Properties:** Min-Heap on values ($val[\text{parent}] \leq val[\text{child}]$) + In-order yields original array indices ($1..n$). Root is global minimum. Subtree of u corresponds to the maximal subarray $[L_u, R_u]$ where $A[u]$ is the minimum ($size(u) = R_u - L_u + 1$).

- $O(n)$ **Stack Build:** Maintain the right spine in a stack. For $i = 1..n$: pop u with $A[u] > A[i]$; last popped becomes $left[i] = u$; remaining top gets $right[\text{top}] = i$. Push i .

- **Key Applications:**

- RMQ $\iff$ LCA: Range minimum $\min_{l \leq k \leq r} A[k] = A[\text{LCA}(l, r)]$.
- Max Rectangle in Histogram: $\max_u (A[u] \cdot size(u))$.
- D&C to Tree DP: Transforms array D&C splitting at minimum into subtree DP.

1.5 Persistent Segment Tree

Complexity: $O(\log n)$ per query/update

Persistent Segment Tree supporting point updates and historic version queries

```

1 struct Node{ //Use Long long if MLE.
2     int sum;
3     //int num;
4     int leftChild;
5     int rightChild;
6
7     Node(int _sum = 0, int _leftChild = 0, int _rightChild = 0){
8         sum = _sum;
9         leftChild = _leftChild;
10        rightChild = _rightChild;
11    }
12 };
13
14 vector<Node> st;
15 vector<int> version;
16
17 void update(int nn, int l, int r, int pos, int val){
18     if (l == r){
19         st[nn].sum = val;
20         return;
21     }
22     int mid = (l + r) >> 1;
23     if (pos <= mid){
24         st.push_back(st[st[nn].leftChild]);
25         st[nn].leftChild = st.size() - 1;
26         update(st[nn].leftChild, l, mid, pos, val);
27     }
28     else{
29         st.push_back(st[st[nn].rightChild]);
30
31         st[nn].rightChild = st.size() - 1;
32         update(st[nn].rightChild, mid + 1, r, pos, val);
33     }
34
35     st[nn].sum = st[st[nn].leftChild].sum + st[st[nn].rightChild].sum;
36 }
37
38 int get(int nn, int l, int r, int u, int v){
39     if (l >= u and r <= v){
40         return st[nn].sum;
41     }
42     if (l > v or r < u) return 0;
43     int mid = (l + r) >> 1;
44     return get(st[nn].leftChild, l, mid, u, v)
45         + get(st[nn].rightChild, mid + 1, r, u, v);
46 }
47
48 ll getKth(int nn, int l, int r, int k){ //Remember to add variable
        "number" to persistent
49     int leftChild = st[nn].leftChild;
50     int rightChild = st[nn].rightChild;
51     int mid = (l + r) >> 1;
52     if (l == r){
53         return value[l] * min(st[nn].num, k * 1LL);
54     }
55     if (st[leftChild].num < k){
56         return st[leftChild].sum + getKth(rightChild, mid + 1, r, k -
            st[leftChild].num);
57     }
58     else return getKth(leftChild, l, mid, k);
59 }
60
61
62 //Use:
63 st.pb(Node(0, 0, 0));
64 version.pb(0);
65 update(version[0], 1, n, i, a[i]);
66
67 { //Add a copy version of the kth version

```

```

68 int k;
69 cin >> k;
70 k--;
71 version.pb(st.size());
72 st.pb(st[version[k]]);
73 }
74 { //Modify and get Range
75 if (type == 1){ //Update pos
76     update(version[k], 1, n, a, b);
77 }
78 else if (type == 2){ //Get Range
79     cout << get(version[k], 1, n, a, b) << endl;
80 }
81 }

```

1.6 Parallel Binary Search

Complexity: $\mathcal{O}((N + Q) \log(\text{range}) \log N)$

Offline simultaneous binary search over multiple monotonic queries

```

1 void PBS(int l, int r, vector<int>& candi) {
2 if(l + 1 == r || candi.empty()) { //trivial case.
3 for(auto c : candi) ans[c] = 1;
4 return;
5 }
6 int mid = (l + r) >> 1;
7 // do things: events from [l, mid)
8
9 // split candi into two parts
10 vector<int> ok, not_ok;
11 for(auto c : candi) {
12 ull sum = 0;
13 //Query information
14
15 if(target[c] <= sum /* Ok Condition */) ok.push_back(c);
16 else {
17     target[c] -= sum; //Reduce the condition
18     not_ok.push_back(c);
19 }
20 }
21 // undo things: events from [l, mid)
22
23 // continue binary search and free memory
24 PBS(l, mid, ok); vector<int> ().swap(ok);
25 PBS(mid, r, not_ok); vector<int> ().swap(not_ok);
26 }

```

1.7 Wavelet Tree

Complexity: $\mathcal{O}(\log(\text{alphabet}))$ per query

Wavelet Tree for range k -th element, range frequency, and range counting in $[L, R]$

```

1 // Complexity:  $\mathcal{O}(N \log \Sigma)$  build,  $\mathcal{O}(\log \Sigma)$  query,  $\mathcal{O}(N \log \Sigma)$  space
2 // Wavelet Tree: Compact structure for range k-th, range count, and pred/succ
3
4 1. Structure & Index Mapping:
5 - Node represents value range [L, R] with mid = (L + R) / 2.
6 - Sequence at node keeps original relative order:
7   Elements <= mid go to Left child, > mid go to Right child.
8 - b[i] = count of elements <= mid in prefix [0..i] (1-based: b[i] = b[i-1] + (A[i] <= mid)).
9 - Index transformation for range [l, r]:
10   Left child range: [b[l - 1] + 1, b[r]]
11   Right child range: [l - b[l - 1], r - b[r]]

```

```

12
13 2. Core Queries:
14 function kth_smallest(node, l, r, k):
15     if L == R: return L
16     c = b[r] - b[l - 1] // count going left
17     if k <= c:
18         return kth_smallest(Left, b[l - 1] + 1, b[r], k)
19     else:
20         return kth_smallest(Right, l - b[l - 1], r - b[r], k - c)
21
22 function count_le(node, l, r, k): // count elements <= k in [l, r]
23     if l > r or k < L: return 0
24     if R <= k: return r - l + 1
25     c = b[r] - b[l - 1]
26     return count_le(Left, b[l - 1] + 1, b[r], k) +
27         count_le(Right, l - b[l - 1], r - b[r], k)
28
29 function count_in_range(l, r, low, high):
30     return count_le(l, r, high) - count_le(l, r, low - 1)
31
32 3. Implementation Tips:
33 - Coordinate Compression: Compress values to [1, distinct_N] so depth is  $\mathcal{O}(\log N)$ .
34 - Bitvector Optimization: Store b[] as 64-bit integers; use __builtin_popcountll for  $\mathcal{O}(1)$  rank.
35 - Swap Adjacent: Swapping A[i] and A[i+1] only updates 1 bit per level in  $\mathcal{O}(\log \Sigma)$ .

```

1.8 Segment Tree Beat

Complexity: $\mathcal{O}(n \log n)$ amortized

Segment Tree Beats: Range Chmin/Chmax, historical min/max, and range sum queries

```

1 // Complexity:  $\mathcal{O}((N + Q) \log N)$  amortized;  $\mathcal{O}((N + Q) \log^2 N)$  with Range Add
2 // Segment Tree Beats (Ji Driver Segment Tree) - Core Ideas & Principles
3
4 1. Core Concept & Motivation:
5 - Target operations: Range Chmin (A[i] = min(A[i], x)), Range Chmax, Range Add, Range Sum.
6 - Key Insight: A range chmin(x) only modifies elements strictly greater than x.
7   By isolating the strictly maximum elements, we can update them in  $\mathcal{O}(1)$  without
8   affecting any other elements in the segment.
9
10 2. Maintained Node State:
11 - max1: Strictly largest value in node range.
12 - max2: Strictly second largest value (max2 < max1; -INF if all values equal).
13 - cnt_max: Number of occurrences of max1 in node range.
14 - sum: Sum of all elements in node range.
15 - (Symmetrically maintain min1, min2, cnt_min if Range Chmax is also needed).
16 - Lazy tags: lazy_add (additive shift), lazy_max (chmin bound).
17
18 3. The Three Update Conditions for Range Chmin(x):
19 - Break Condition (Outside or No-Op):
20   Node interval is disjoint from query interval, OR x >= max1.
21   Action: Return immediately (no value in subtree can change).
22 - Tag Condition (Direct  $\mathcal{O}(1)$  Update):
23   Node interval is fully inside query interval, AND max2 < x < max1.
24   Action: Only the cnt_max elements equal to max1 decrease to x.

```

```

25     sum -= cnt_max * (max1 - x); max1 = x; set lazy tag;
26     return.
27 - Recursion Condition (Multiple Values Affected):
28   Node interval is fully inside query interval, BUT x <= max2.
29   Action: Both max1 and max2 (and potentially others) change.
30   Push down pending tags, recurse into both children, then pull/merge.
31
32 4. Maintenance Rules (Pull & Push):
33 - Pull (Merge from children L, R):
34   * sum = L.sum + R.sum
35   * max1 = max(L.max1, R.max1)
36   * cnt_max = sum of cnt_max of children having child.max1 == parent.max1.
37   * max2 = largest value among {L.max1, L.max2, R.max1, R.max2} strictly less than max1.
38 - Push Down (Lazy Propagation):
39   * Additive tags apply to all elements (shift sum, max1, max2, min1, min2).
40   * Chmin tag u.max1 only pushes to children with child.max1 > u.max1:
41     child.sum -= child.cnt_max * (child.max1 - u.max1), child.max1 = u.max1.
42
43 5. Complexity & Potential Analysis:
44 - Define potential Phi = total number of distinct values across all nodes in tree.
45 - Each recursion into the 3rd case (x <= max2) forces at least two distinct values
46   to merge into one, strictly decreasing Phi by >= 1.
47 - Chmin/Chmax + Sum: amortized  $\mathcal{O}((N + Q) \log N)$ .
48 - Range Add creates at most  $\mathcal{O}(\log N)$  new distinct values -> amortized  $\mathcal{O}((N + Q) \log^2 N)$ .
49
50 6. Common Extensions:
51 - Range Divide (A[i] = floor(A[i] / d)): Convert to Range Add when
52   max1 - floor(max1 / d) == min1 - floor(min1 / d), otherwise recurse.
53 - Range Modulo (A[i] = A[i] % d): Break if max1 < d; otherwise recurse
54   (each recursion strictly halves values >= d).
55 - Historical Min/Max/Sum: Maintain historical max tag alongside current tag.

```

1.9 Kd Tree

Complexity: $\mathcal{O}(\sqrt{n})$ per query

K-Dimensional Tree for k -NN nearest neighbor and orthogonal range queries in 2D/3D

```

1 struct point{
2 double dim[2];
3 bool operator != (const point &other){
4     return !(dim[0] == other.dim[0] and dim[1] == other.dim[1]);
5 }
6 } pt[MAXN], tmp[MAXN];
7 struct cmp_point{
8 int axis;
9 cmp_point(int x = 0): axis(x) {}
10 bool operator () (const point &a, const point &b){
11     return a.dim[axis] < b.dim[axis];
12 }
13 };
14 struct kd_tree{
15 int axis;
16 double value;
17 point p;

```

```

18 kd_tree *left ,*right;
19 };
20 typedef kd_tree* node_ptr;
21 node_ptr node;
22 double distance_squared(point a, point b){
23 double sum = 0;
24 for(int i = 0; i < 2; i++){
25     sum += (a.dim[i] - b.dim[i]) * (a.dim[i] - b.dim[i]);
26 }
27 return sqrt(sum);
28 }
29 void build(node_ptr &node, int l, int r, int axis){
30 if (l > r){
31     node = NULL;
32     return;
33 }
34 node = new kd_tree();
35 if (l == r){
36     node->p = pt[l];
37     node->left = node->right = NULL;
38     node->axis = axis;
39     return;
40 }
41 int mid = (l + r) >> 1;
42 nth_element(pt + l, pt + mid, pt + 1 + r, cmp_point(axis));
43 node->value = pt[mid].dim[axis]; node->axis = axis; node->p =
    pt[mid];
44 build(node->left, l, mid - 1, axis ^ 1);
45 build(node->right, mid + 1, r, axis ^ 1);
46 }
47 void NearestNeighbour(node_ptr node, point q, double &ans){
48 if (node == NULL) return;
49 if (q != node->p) minimize(ans, distance_squared(node->p, q));
50 if (node->left == NULL and node->right == NULL){
51     return;
52 }
53 int axis = node->axis;
54 double value = node->value;
55 if (q.dim[axis] <= value){
56     NearestNeighbour(node->left, q, ans);
57     if (q.dim[axis] + ans >= value + eps) NearestNeighbour(node->
        right, q, ans);
58 }
59 else{
60     NearestNeighbour(node->right, q, ans);
61     if (q.dim[axis] - ans <= value - eps) NearestNeighbour(node->
        left, q, ans);
62 }
63 }

```

1.10 BIT2D

Complexity: $\mathcal{O}(\log^2 n)$ per query
 2D Binary Indexed Tree (Fenwick) for 2D range sum queries and point updates

```

1 int BIT2D[1003][1003];
2 void update(int x, int y, int val){
3     for(int i = x; i <= n; i += i & (-i)){
4         for(int j = y; j <= m; j += j & (-j)){
5             BIT2D[i][j] += val;
6         }
7     }
8 }
9
10 int get(int x, int y){
11     int res = 0;
12     for(int i = x; i > 0; i -= i & (-i)){

```

```

13         for(int j = y; j > 0; j -= j & (-j)){
14             res += BIT2D[i][j];
15         }
16     }
17     return res;
18 }

```

1.11 Treap

Complexity: $\mathcal{O}(\log n)$ expected per operation
 Randomized Binary Search Tree supporting array split/merge and implicit key operations

```

1 int RAND(){
2 int res = 0;
3 FOR(i, 1, 30){
4     res += (1LL << (rand() % 30));
5 }
6 return res;
7 }
8 struct TreapNode{
9 TreapNode *leftChild, *rightChild;
10 int val = 0, sum = 0, weight;
11 int lazy = 0, sz = 0;
12
13 TreapNode(int _val = 0){
14     val = _val;
15     sum = val;
16     weight = RAND();
17     lazy = 0;
18     sz = 1;
19     leftChild = rightChild = nullptr;
20 }
21 };
22 typedef TreapNode* Treap;
23 int getSum(Treap root){
24 if (root == nullptr) return 0;
25 return root->sum;
26 }
27 int getSize(Treap root){
28 if (root == NULL) return 0;
29 return root->sz;
30 }
31 void update(Treap &root){
32 if (root == NULL) return;
33 root->sum = root->val + getSum(root->leftChild) + getSum(root->
    rightChild);
34 root->sz = 1 + getSize(root->leftChild) + getSize(root->rightChild)
    ;
35 }
36 void fix(Treap root){
37 if (root == NULL) return;
38 if (root->lazy){
39     swap(root->leftChild, root->rightChild);
40     if (root->leftChild != NULL) root->leftChild->lazy ^= 1;
41     if (root->rightChild != NULL) root->rightChild->lazy ^= 1;
42 }
43 root->lazy = 0;
44 }
45 void split(Treap rt, Treap &l, Treap &r, int key, int add){
46
47 if (rt == NULL) {
48     l = r = NULL;
49     return;
50 }
51 fix(rt);
52
53 int imkey = add + getSize(rt->leftChild);

```

```

54 if (imkey <= key){
55     l = rt;
56     split(rt->rightChild, l->rightChild, r, key, imkey + 1);
57     update(rt);
58 }
59 else{
60     r = rt;
61     split(rt->leftChild, l, r->leftChild, key, add);
62     update(rt);
63 }
64 }
65
66 void merge(Treap &rt, Treap l, Treap r){
67 fix(l);
68 fix(r);
69 if (!l or !r){
70     rt = ((!l) ? r : l);
71     return;
72 }
73
74 if (l->weight < r->weight){
75     rt = r;
76     merge(rt->leftChild, l, r->leftChild);
77     update(rt);
78 }
79 else{
80     rt = l;
81     merge(rt->rightChild, l->rightChild, r);
82     update(rt);
83 }
84 }
85
86 void go(Treap rt){
87 if (rt == NULL) return;
88
89 go(rt->leftChild);
90 cout << rt->val << ' ';
91
92 go(rt->rightChild);
93
94 }
95 Treap rt;

```

1.12 Splay Tree / Link - cut Tree

Complexity: $\mathcal{O}(\log n)$ amortized per operation
 Link-Cut Tree supporting dynamic path queries, link, cut, and re-root on forests

```

1 struct Node {
2 int x;
3 Node *l = 0;
4 Node *r = 0;
5 Node *p = 0;
6 bool rev = false;
7 Node() = default;
8 Node(int v) { x = v; }
9 void push() {
10 if (rev) {
11     rev = false;
12     swap(l, r);
13     if (l) l->rev ^= true;
14     if (r) r->rev ^= true;
15 }
16 }
17 bool is_root() { return p == 0 || (p->l != this && this != p->r); }
18 };
19 struct LCT {

```

```

20 vector<Node> a;
21 LCT(int n) {
22     a.resize(n + 1);
23     for (int i = 1; i <= n; ++i) a[i].x = i;
24 }
25 void rot(Node *c) {
26     auto p = c->p;
27     auto g = p->p;
28     if (!p->is_root()) (g->r == p ? g->r : g->l) = c;
29     p->push();
30     c->push();
31     if (p->l == c) { // rtl
32         p->l = c->r;
33         c->r = p;
34         if (p->l) p->l->p = p;
35     } else { // rtr
36         p->r = c->l;
37         c->l = p;
38         if (p->r) p->r->p = p;
39     }
40     p->p = c;
41     c->p = g;
42 }
43 void splay(Node *c) {
44     while (!c->is_root()) {
45         auto p = c->p;
46         auto g = p->p;
47         if (!p->is_root()) rot((g->r == p) == (p->r == c) ? p : c);
48         rot(c);
49     }
50     c->push();
51 }
52 Node *access(int v) {
53     Node *last = 0;
54     Node *c = &a[v];
55     for (Node *p = c; p; p = p->p) {
56         splay(p);
57         p->r = last;
58         last = p;
59     }
60     splay(c);
61     return last;
62 }
63 void make_root(int v) {
64     access(v);
65     auto *c = &a[v];
66     if (c->l) c->l->rev ^= true, c->l = 0;
67 }
68 void link(int u, int v) {
69     make_root(u);
70     Node *c = &a[u];
71     c->p = &a[v];
72 }
73 void cut(int u) {
74     access(u);
75     assert(a[u].l);
76     a[u].l->p = 0;
77     a[u].l = 0;
78 }
79 int lca(int u, int v) {
80     if (u == v) return u;
81     access(u); access(v);
82     if (a[u].p == 0) return -1;
83     splay(&a[u]);
84     return (a[u].p ? a[u].p->x : a[u].x);
85 }
86 };

```

1.13 Farach - Colton - Bender

Complexity: $\mathcal{O}(n)$ build, $\mathcal{O}(1)$ query
 $\langle \mathcal{O}(n), \mathcal{O}(1) \rangle$ LCA via Euler Tour ± 1 RMQ and Four Russians speedup

```

1 // Complexity:  $\mathcal{O}(N)$  preprocessing,  $\mathcal{O}(1)$  query,  $\mathcal{O}(N)$  space
2 // Lowest Common Ancestor (LCA) via  $\pm 1$  RMQ & Four Russians
   Speedup
3
4 1. Reduction (Tree LCA  $\rightarrow \pm 1$  RMQ):
5   - Run Euler Tour DFS: record traversal order E[] and depth D[]
     of length  $M = 2N - 1$ .
6   - first[u] = index of first occurrence of node u in E[].
7   - LCA(u, v) = E[ argmin_{i in [first[u], first[v]]} D[i] ].
8   - Property:  $|D[i] - D[i - 1]| = 1$  for all adjacent steps ( $\pm 1$  RMQ).
9
10 2. Block Partitioning & Sparse Table:
11   - Block size  $B = \max(1, \text{floor}(\log_2(M) / 2))$ , number of blocks  $K$ 
     =  $\text{ceil}(M / B)$ .
12   - Compute minimum index of each block: block_min[0..K-1].
13   - Build Sparse Table over block_min[] in  $\mathcal{O}(K \log K) = \mathcal{O}(N)$  time
     and space.
14
15 3. In-Block Precomputation via Bitmasks (Four Russians):
16   - For block b, bit (j - 1) = 1 if  $D[b*B + j] > D[b*B + j - 1]$ 
     else 0.
17   - Only  $2^{(B - 1)} \leq \sqrt{M}$  unique block profiles exist.
18   - Precompute all-pairs RMQ table of size  $B \times B$  for each unique
     mask in  $\mathcal{O}(B^2)$ .
19   - Total table precomputation:  $\mathcal{O}(2^B * B^2) = \mathcal{O}(\sqrt{N} \log^2 N)$ 
     =  $\mathcal{o}(N)$ .
20
21 4. Query LCA / RMQ in  $\mathcal{O}(1)$ :
22     function query_rmq(L, R):
23         bL = L / B, bR = R / B
24         if bL == bR:
25             return in_block_rmq(bL, L % B, R % B)
26         ans = min_by_depth(in_block_rmq(bL, L % B, B - 1),
27                             in_block_rmq(bR, 0, R % B))
28         if bL + 1 < bR:
29             ans = min_by_depth(ans, sparse_table_query(bL + 1, bR -
30                                                         1))
31         return ans
32
33     function query_lca(u, v):
34         L = min(first[u], first[v]), R = max(first[u], first[v])
35         return E[query_rmq(L, R)]

```

1.14 Trie

Complexity: $\mathcal{O}(\text{length})$ per query
 Binary/Alphabet Prefix Tree for dictionary lookup and XOR max queries

```

1 //Vector Implementation
2 struct TrieNode{
3     int child[2];
4     int sz = 0;
5     TrieNode(){
6         memset(child, -1, sizeof(child));
7         sz = 0;
8     }
9 };
10
11 const int B = 30;
12 struct Trie{
13     vector<TrieNode> T = {TrieNode()};

```

```

14
15 void addNum(int num, int s){
16     int p = 0;
17     for(int i = B; i >= 0; i--){
18         int b = (num >> i)&1;
19         if (T[p].child[b] == -1){
20             T[p].child[b] = T.size();
21             T.pb(TrieNode());
22         }
23         T[p].sz += s;
24         p = T[p].child[b];
25     }
26     T[p].sz += s;
27 }
28
29 bool exist(int num){
30     int p = 0;
31     for(int i = B; i >= 0; i--){
32         int b = (num >> i)&1;
33         if (T[p].child[b] == -1) return false;
34         p = T[p].child[b];
35     }
36     return T[p].sz;
37 }
38
39 int getMinXor(int num){
40     int res = 0;
41     int p = 0;
42     for(int i = B; i >= 0; i--){
43         int b = (num >> i)&1;
44         int c1 = T[p].child[b];
45         int c2 = T[p].child[!b];
46
47         if (c1 != -1 and T[c1].sz > 0) p = c1;
48         else if (c2 != -1 and T[c2].sz > 0) {
49             p = c2;
50             res += (1LL << i);
51         }
52     }
53     return res;
54 }
55 };
56
57 //-----
58 //Pointer Implementation
59 struct Trie
60 {
61     struct Node
62     {
63         Node *child[26];
64         int cnt, is_end;
65         Node()
66         {
67             for (int i = 0; i < 26; i++) child[i] = 0;
68             cnt = is_end = 0;
69         }
70     };
71     Node *root, *cur;
72     Trie()
73     {
74         root = new Node();
75     }
76
77     void add(string s)
78     {
79         cur = root;
80         for (char i : s)
81         {
82             if (cur->child[i - 'a'] == NULL) cur->child[i - 'a'] = new

```

```

    Node();
83     cur = cur->child[i - 'a'];
84     cur->cnt++;
85 }
86 cur->is_end++;
87 }
88
89 bool check(string s)
90 {
91     cur = root;
92     for (char i : s)
93     {
94         if (cur->child[i - 'a'] == NULL) return(0);
95         cur = cur->child[i - 'a'];
96     }
97     return(cur->is_end);
98 }
99
100 int cnt(string s)
101 {
102     cur = root;
103     for (char i : s)
104     {
105         if (cur->child[i - 'a'] == NULL) return(0);
106         cur = cur->child[i - 'a'];
107     }
108     return(cur->cnt);
109 }
110
111 void del(string s)
112 {
113     stack <Node*> st;
114     cur = root;
115     for (int i = s.size() - 1; i >= 0; i--)
116     {
117         st.push(cur);
118         cur = cur->child[s[i] - 'a'];
119         cur->cnt--;
120     }
121     st.push(cur);
122     for (int i = 0; i < s.size(); i++)
123     {
124         if (st.top()->cnt == 0)
125         {
126             delete(st.top());
127             st.pop();
128             st.top()->child[s[i] - 'a'] = NULL;
129         }
130     }
131 }
132 };

```

1.15 Lichao Tree

Complexity: $\mathcal{O}(\log n)$ per query/insertion
Li Chao Segment Tree for online dynamic line insertions and minimum/maximum evaluation

```

1  const int maxn = 2e5;
2
3  point line[4 * maxn];
4
5  void add_line(point nw, int v = 1, int l = 0, int r = maxn) {
6      int m = (l + r) / 2;
7      bool lef = f(nw, l) < f(line[v], l);
8      bool mid = f(nw, m) < f(line[v], m);
9      if(mid) {
10         swap(line[v], nw);

```

```

11     }
12     if(r - l == 1) {
13         return;
14     } else if(lef != mid) {
15         add_line(nw, 2 * v, l, m);
16     } else {
17         add_line(nw, 2 * v + 1, m, r);
18     }
19 }

```

1.16 Line Container

Complexity: $\mathcal{O}(\log n)$ amortized per operation
Convex Hull of dynamic lines with arbitrary slopes queried at arbitrary x

```

1  ///// line container start (max query) /////
2  struct Line {
3      mutable ll k, m, p;
4      bool operator<(const Line& o) const { return k < o.k; }
5      bool operator<(ll x) const { return p < x; }
6  }; // (for doubles, use inf = 1/.0, div(a,b) = a/b)
7  struct LineContainer : multiset<Line, less<>> {
8      static const ll inf = LLONG_MAX; // div: floor
9      ll div(ll a, ll b) { return a / b - ((a ^ b) < 0 && a % b); }
10     bool isect(iterator x, iterator y) {
11         if (y == end()) return x->p = inf, 0;
12         if (x->k == y->k) x->p = x->m > y->m ? inf : -inf;
13         else x->p = div(y->m - x->m, x->k - y->k);
14         return x->p >= y->p;
15     }
16     void add(ll k, ll m) {
17         auto z = insert({k, m, 0}), y = z++, x = y;
18         while (isect(y, z)) z = erase(z);
19         if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
20         while ((y = x) != begin() && (--x)->p >= y->p) isect(x,
21             erase(y));
22         ll query(ll x) { assert(!empty());
23             auto l = *lower_bound(x); return l.k * x + l.m; }
24 };

```

1.17 Arpa Trick

Complexity: $\mathcal{O}(n\alpha(n))$
Fast offline range maximum/minimum query via DSU

```

1  Offline RMQ in  $\mathcal{O}(N \cdot \alpha(N))$  using DSU and Monotonic Stack.
2
3  Idea:
4  1. Group queries by their right endpoint  $R$ .
5  2. Iterate  $R$  from  $1$  to  $N$ , maintaining a monotonic stack of
   indices of strictly decreasing elements.
6  3. When the current element  $a[R]$  pops an index  $u$  from the stack
   (because  $a[u] \leq a[R]$ ), we merge  $u$  into  $R$  in a DSU by
   setting  $parent[u] = R$ .
7  4. After updating the stack and DSU for  $R$ , to answer a query  $[L,
   R]$ :
8     - The index of the maximum element in  $[L, R]$  is exactly '
   find_root(L)'.
9
10 Why it works: 'find_root(L)' follows the chain of "next greater
   elements" and stops at the largest element  $\leq R$  (since
   elements  $\geq R$  haven't been processed yet).

```

1.18 Blocky Data Structure

Complexity: $\mathcal{O}(n\sqrt{n})$
Square root decomposition: Block Array and Hilbert-ordered Mo's algorithm

```

1
2 //Sqrt Decomposition DS
3 //Query:  $\mathcal{O}(1)$ 
4 //Update:  $\mathcal{O}(\sqrt{n})$ 
5
6
7 struct BlockyArray{
8     vector<int> arr, lazy; //zero based
9     BlockyArray(int n = 0){
10         arr.resize(n + 9, 0);
11         lazy.resize(n + 9, 0);
12     }
13     void updateUp(int pos, int val){
14         pos++;
15         while(pos < arr.size() and pos % block != 0) arr[pos++] +=
16             val;
17         for(int i = pos; i < arr.size(); i += block) lazy[i /
18             block] += val;
19     }
20     void updateDown(int pos, int val){ //one based
21         //cap nhất nhưng gia tri nho hon
22         pos
23         pos--;
24         while(pos >= 0 and pos % block != block - 1) arr[pos--] +=
25             val;
26         for(int i = pos; i >= 0; i -= block) lazy[i / block] +=
27             val;
28     }
29     int get(int pos){
30         return arr[pos] + lazy[pos / block];
31     }
32 };
33
34 //MO - Hilbert
35 inline int64_t hilbertOrder(int x, int y, int pow, int rotate) {
36     if (pow == 0) { return 0; }
37     int hpow = 1 << (pow-1);
38     int seg = (x < hpow) ? ((y < hpow) ? 0 : 3) : ((y < hpow) ? 1 : 2
39         );
40     seg = (seg + rotate) & 3;
41     const int rotateDelta[4] = {3, 0, 0, 1};
42     int nx = x & (x ^ hpow), ny = y & (y ^ hpow);
43     int nrot = (rotate + rotateDelta[seg]) & 3;
44     int64_t subSquareSize = int64_t(1) << (2*pow - 2);
45     int64_t ans = seg * subSquareSize;
46     int64_t add = hilbertOrder(nx, ny, pow-1, nrot);
47     ans += ((seg == 1 || seg == 2) ? add : (subSquareSize - add - 1))
48         ;
49     return ans;
50 }
51
52 struct Query{
53     int l, r, id, ord;
54     Query(int _l, int _r, int _id){
55         l = _l; r = _r; id = _id;
56         ord = hilbertOrder(l, r, 17, 0);
57     }
58     bool operator < (const Query &other) const{ return ord < other.
59         ord;}
60 };
61
62 vector<Query> query;
63 void pseudo_mo(int lo, int hi, int id){
64     while(r < hi){

```

```

56     sweeplineM0[l - 1].pb(Segment(r + 1, hi, -1, id));
57     r = hi;
58 }
59 while(l > lo){
60     sweeplineM0[r].pb(Segment(lo, l - 1, 1, id));
61     l = lo;
62 }
63 while(r > hi){
64     sweeplineM0[l - 1].pb(Segment(hi + 1, r, 1, id));
65     r = hi;
66 }
67 while(l < lo){
68     sweeplineM0[r].pb(Segment(l, lo - 1, -1, id));
69     l = lo;
70 }
71 }

```

1.19 Ultimate Hashmap

Complexity: $\mathcal{O}(1)$ expected per lookup

Fast custom hash map with custom splitmix64 hash function

```

1 // NOTE: #include <ext/pb_ds/assoc_container.hpp> MUST be placed
  BEFORE #define int long long
2 #include <ext/pb_ds/assoc_container.hpp>
3 using namespace __gnu_pbds;
4
5 struct custom_hash {
6     static uint64_t splitmix64(uint64_t x) {
7         x += 0x9e3779b97f4a7c15;
8         x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
9         x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
10        return x ^ (x >> 31);
11    }
12
13    size_t operator()(uint64_t x) const {
14        static const uint64_t FIXED_RANDOM = chrono::steady_clock::
15            now().time_since_epoch().count();
16        return splitmix64(x + FIXED_RANDOM);
17    };
18 gp_hash_table<int, int, custom_hash> table;

```

1.20 Erasable Priority Queue

Complexity: $\mathcal{O}(\log n)$ per operation

Max/Min Priority Queue with lazy removal of arbitrary elements

```

1 template<class T=int, class O=less<T>>
2 struct pq_set {
3     priority_queue<T, vector<T>, O> q, del;
4     const T& top() const { return q.top(); }
5     int size() const { return (int)q.size()-del.size(); }
6     bool empty() const { return !size(); }
7     void insert(const T x) { q.push(x); flush(); }
8     void pop() { q.pop(); flush(); }
9     void erase(const T x) { del.push(x); flush(); }
10    void flush() { while(del.size() && q.top()==del.top())
11        q.pop(), del.pop(); }
12 };

```

1.21 Expandable Data Structure

Complexity: $\mathcal{O}(A \log n)$ amortized insertion, for static structure buildable in $\mathcal{O}(A)$

Binary doubling technique: converts static data structures (e.g. Aho, CHT) to support online insertions

```

1 void Insert(){ //Insert something here.
2 for(int i = 0; i < 20; i++){
3     if (block[i].size() > (1 << i)){
4         //Push the element onto the next set
5     }
6     else{
7         //rebuild the data structure
8         break;
9     }
10 }
11 }
12 void query(){
13 for(int i = 0; i < 20; i++){
14     //query on each block
15 }
16 }

```

1.22 Interval Set

Complexity: $\mathcal{O}(\log n)$ amortized per operation

Disjoint interval set (ODT style): maintains non-overlapping intervals, range assignment, and adjacent merging based on predicate.

```

1 struct Node {
2     int l, r; char c;
3     bool operator<(const Node& o) const { return l < o.l; }
4 };
5 struct IntervalSet : set<Node> {
6     bool can_merge(char a, char b) const { return a == b; }
7
8     auto assign(int l, int r, char c) {
9         vector<Node> rem, add;
10
11         // 1. Find first interval intersecting with [l, r]
12         auto it = upper_bound({l, (int)2e9, 0});
13         if (it != begin() && prev(it)->r >= l) --it;
14
15         // 2. Erase intersecting intervals & preserve leftovers at
16         // both ends
17         while (it != end() && it->l <= r) {
18             auto [L, R, C] = *it;
19             rem.push_back(*it);
20             it = erase(it);
21             if (L < l) add.push_back(*insert({L, l - 1, C}).first);
22             if (R > r) add.push_back(*insert({r + 1, R, C}).first);
23         }
24
25         // 3. Merge with neighbors if can_merge holds
26         it = lower_bound({l, 0, 0});
27         if (it != begin() && prev(it)->r == l - 1 && can_merge(prev(it)->c, c)) {
28             l = prev(it)->l; rem.push_back(*prev(it)); erase(prev(it));
29         }
30         if (it != end() && it->l == r + 1 && can_merge(it->c, c)) {
31             r = it->r; rem.push_back(*it); erase(it);
32         }
33
34         // 4. Insert new interval
35         add.push_back(*insert({l, r, c}).first);
36         return pair{rem, add};
37 };

```

2 Graph

2.1 Minimum Cost Flow

Complexity: $\mathcal{O}(F \cdot m \log n)$

Min-cost K Flow: Transfer k packages through graph with capacities and costs

```

1 // Complexity:  $\mathcal{O}(F * M \log N)$  using SPFA / Dijkstra with node
  potentials
2 // Finds min cost to send F units of flow from source s to sink t
3
4 struct Edge:
5     to, rev, cap, flow = 0, cost
6
7 function min_cost_flow(s, t, max_flow_target):
8     pot[N] = 0, total_flow = 0, total_cost = 0
9     // Compute initial potential with SPFA if negative costs exist
10
11     while total_flow < max_flow_target:
12         priority_queue pq // (dist, u)
13         dist[0..N-1] = INF, parent_edge[0..N-1] = -1
14         dist[s] = 0, pq.push({0, s})
15
16         while not pq.empty():
17             d, u = pq.pop()
18             if d > dist[u]: continue
19             for e in adj[u]:
20                 if e.cap - e.flow > 0:
21                     reduced_cost = e.cost + pot[u] - pot[e.to]
22                     if dist[e.to] > dist[u] + reduced_cost:
23                         dist[e.to] = dist[u] + reduced_cost
24                         parent_edge[e.to] = e
25                         pq.push({dist[e.to], e.to})
26
27             if dist[t] == INF: break // No more augmenting path
28             for i = 0 to N - 1: pot[i] += dist[i]
29
30             push_flow = max_flow_target - total_flow
31             v = t
32             while v != s:
33                 e = parent_edge[v]
34                 push_flow = min(push_flow, e.cap - e.flow)
35                 v = adj[e.to][e.rev].to
36
37             v = t
38             while v != s:
39                 e = parent_edge[v]
40                 e.flow += push_flow
41                 adj[e.to][e.rev].flow -= push_flow
42                 total_cost += push_flow * e.cost
43                 v = adj[e.to][e.rev].to
44             total_flow += push_flow
45
46     return (total_flow, total_cost)

```

2.2 MinCut - MaxFlow

Complexity: $\mathcal{O}(EV^2)$

Dinic: Max number of paths / vertex-disjoint paths from source to sink.

- Grid 2-Coloring:** Partition cells by $(r + c) \bmod 2 \implies$ Bipartite graph. Used for domino tiling, non-attacking pieces, laser grid beam shooting.

- Vertex Splitting** ($u_{in} \rightarrow u_{out}$): Restricts vertex capacity or forces vertex visit (capacity c_u , cost w_u).

- Max-Weight Closure (Project Selection):**

- Rule: If choosing item u , must also choose item v (directed edge $u \rightarrow v$ with cap = ∞).
- For positive weight $W_u > 0$: Add edge $S \rightarrow u$ with cap = W_u .
- For negative weight $W_v < 0$: Add edge $v \rightarrow T$ with cap = $-W_v$.
- Max Profit = $\sum_{W_u > 0} W_u - \text{MinCut}(S, T)$.

• **Two-Choice Binary Partitioning:** Each item i assigned to set S (cost A_i) or T (cost B_i). If $i \in S, j \in T$, penalty $C_{i,j}$. Add $S \rightarrow i (B_i), i \rightarrow T (A_i), i \rightarrow j (C_{i,j})$. Total cost = Min-Cut.

• **Circulation with Demands** $[L_e, R_e]$: Replace edge $e = (u, v)$ with cap $R_e - L_e$. Let $d(u) = \sum_{in} L - \sum_{out} L$. Add super-source S^* , super-sink T^* . For $d(u) > 0$, add $S^* \rightarrow u$ with cap $d(u)$; for $d(u) < 0$, add $u \rightarrow T^*$ with cap $-d(u)$. Feasible $\iff$ all S^* edges saturated.

```

1 // Complexity: O(V^2 * E) general, O(E * sqrt(V)) for unit networks
  / bipartite matching
2 // Dinic's Algorithm: Repeated BFS level graph construction and DFS
  blocking flow
3
4 struct Edge:
5     to, rev, cap, flow = 0
6
7 int level[N], ptr[N]
8
9 function bfs_level(s, t):
10     fill(level, -1); level[s] = 0
11     queue Q; Q.push(s)
12     while not Q.empty():
13         u = Q.pop()
14         for e in adj[u]:
15             if e.cap - e.flow > 0 and level[e.to] == -1:
16                 level[e.to] = level[u] + 1
17                 Q.push(e.to)
18     return level[t] != -1
19
20 function dfs_blocking(u, t, pushed):
21     if pushed == 0 or u == t: return pushed
22     for cid = ptr[u] to len(adj[u]) - 1:
23         ptr[u] = cid; e = adj[u][cid]
24         if level[e.to] == level[u] + 1 and e.cap - e.flow > 0:
25             tr = dfs_blocking(e.to, t, min(pushed, e.cap - e.flow))
26             if tr == 0: continue
27             e.flow += tr; adj[e.to][e.rev].flow -= tr
28             return tr
29     return 0
30
31 function dinic_max_flow(s, t):
32     flow = 0
33     while bfs_level(s, t):
34         fill(ptr, 0)
35         while pushed = dfs_blocking(s, t, INF):
36             flow += pushed
37     return flow

```

2.3 Bipartite Matching

Complexity: $\mathcal{O}(E\sqrt{V})$

Hopcroft-Karp: Node Cover, Node-disjoint path cover, General path cover, Antichain (Dilworth).

• **König's Theorem:** Max Matching = Min Vertex Cover.

• **Max Independent Set:** $|V| - \text{Min Vertex Cover} = |V| - \text{Max Matching}$.

• **Min Edge Cover** (no isolated vertices): $|V| - \text{Max Matching}$.

• **Min Path Cover on DAG:**

– *Vertex-disjoint:* Split $u \rightarrow u_{out}, u_{in}$; build bipartite matching $u_{out} \rightarrow v_{in}$ for edge $u \rightarrow v$. Result: $|V| - \text{Max Matching}$.

– *General (paths can share vertices):* Compute transitive closure (Floyd-Warshall / reachability), then solve vertex-disjoint path cover.

• **Dilworth's Theorem:** In any finite poset (DAG reachability), the length of the **Maximum Antichain** (largest subset of mutually unreachable vertices) equals the **Minimum Path Cover**.

• **Hall's Marriage Theorem:** Bipartite (X, Y) has a matching saturating $X \iff \forall S \subseteq X, |N(S)| \geq |S|$.

• **Max Matching Size via Hall:** $|X| - \max_{S \subseteq X} (|S| - |N(S)|)$.

```

1 // Complexity: O(E * sqrt(V))
2 // Maximum Bipartite Matching on Left set L [1..n] and Right set R
  [1..m]
3
4 int pair_u[MAXN], pair_v[MAXM], dist[MAXN]
5
6 function bfs_hopcroft():
7     queue Q
8     for u = 1 to n:
9         if pair_u[u] == 0: dist[u] = 0; Q.push(u)
10        else: dist[u] = INF
11    dist[0] = INF
12    while not Q.empty():
13        u = Q.pop()
14        if dist[u] < dist[0]:
15            for v in adj[u]:
16                if dist[pair_v[v]] == INF:
17                    dist[pair_v[v]] = dist[u] + 1
18                    Q.push(pair_v[v])
19    return dist[0] != INF
20
21 function dfs_hopcroft(u):
22     if u != 0:
23         for v in adj[u]:
24             if dist[pair_v[v]] == dist[u] + 1 and dfs_hopcroft(
                pair_v[v]):
25                 pair_v[v] = u; pair_u[u] = v
26                 return True
27     dist[u] = INF
28     return False
29     return True
30
31 function max_matching():
32     fill(pair_u, 0); fill(pair_v, 0); matching = 0
33     while bfs_hopcroft():
34         for u = 1 to n:
35             if pair_u[u] == 0 and dfs_hopcroft(u):
36                 matching += 1
37     return matching

```

2.4 Hungarian Matching

Complexity: $\mathcal{O}(V^3)$

Min-cost perfect assignment on bipartite graph

```

1 // Complexity: O(N^3) time, O(N^2) space
2 // Minimum weight perfect matching on 1-indexed N x M bipartite
  graph (N <= M)
3
4 function hungarian(cost[1..N][1..M]):
5     u[0..N] = 0, v[0..M] = 0, p[0..M] = 0, way[0..M] = 0
6     for i = 1 to N:
7         p[0] = i, j0 = 0
8         minv[0..M] = INF, used[0..M] = False

```

```

9     repeat:
10         used[j0] = True
11         i0 = p[j0], delta = INF, j1 = 0
12         for j = 1 to M:
13             if not used[j]:
14                 cur = cost[i0][j] - u[i0] - v[j]
15                 if cur < minv[j]: minv[j] = cur; way[j] = j0
16                 if minv[j] < delta: delta = minv[j]; j1 = j
17         for j = 0 to M:
18             if used[j]: u[p[j]] += delta; v[j] -= delta
19             else: minv[j] -= delta
20         j0 = j1
21         until p[j0] == 0
22     repeat:
23         j1 = way[j0]; p[j0] = p[j1]; j0 = j1
24         until j0 == 0
25     // Result: matched left node of right node j is p[j], min cost
    is -v[0]
26     return -v[0]

```

2.5 General Matching

Complexity: $\mathcal{O}(V^3)$

Edmonds Blossom: Maximum Matching on general graphs.

• **Tutte Matrix & Rabin-Vazirani** ($\mathcal{O}(V^3)$): For $(u, v) \in E$ ($u < v$), set $T_{u,v} = r \in [1, p-1]$, $T_{v,u} = -r \pmod{p}$. Size = $\frac{1}{2} \text{rank}(T)$ (Gauss mod p).

– *Rabin-Vazirani Reconstruction:* Compute $A = T^{-1} \pmod{p}$ (on $2K \times 2K$ basis). Edge (u, v) is in matching $\iff A_{u,v} \neq 0$. Pick (u, v) with $A_{u,v} \neq 0$, match them, then update A in $\mathcal{O}(V^2)$ via Schur complement: $A'_{x,y} = A_{x,y} + \frac{A_{x,u}A_{y,v} - A_{x,v}A_{y,u}}{A_{u,v}} \pmod{p}$.

Repeat for all K pairs $\implies \mathcal{O}(V^3)$ total.

• **Max b -Matching Reduction** ($\deg(i) \leq a_i$): Đỉnh $i \rightarrow a_i$ slot ($a_i \leftarrow \min(a_i, \deg(i))$). Cạnh $e = (u, v) \rightarrow L_e, R_e$, nối $L_e - R_e$, $L_e - \text{slot}(u)$, $R_e - \text{slot}(v)$. Bỏ e : match $L_e - R_e$ (+1). Chọn e : match L_e, R_e với slot (+2, gain +1). Aux $|V'| \leq 4M \implies$ Blossom.

```

1 // TUTTE MATRIX & RABIN-VAZIRANI MATCHING - O(V^3)
2 // -----
3 // p is a large prime (1e9 + 7). All operations mod p.
4
5 function build_tutte_matrix(G):
6     T = zeros(N, N)
7     for each edge (u, v) in G with u < v:
8         r = random(1, p - 1)
9         T[u][v] = r, T[v][u] = p - r
10    return T
11
12 // Symmetric Gaussian Elimination -> Basis S (|S| = 2K)
13 function find_basis(T):
14     M = copy(T), S = empty_list
15     for i = 0 to N - 1:
16         if i in S: continue
17         find pivot j > i (j not in S and M[i][j] != 0)
18         if not found: continue
19         add i, j to S
20         inv = (M[i][j])^(-1) mod p
21         for r = 0 to N - 1 not in S:
22             for c = 0 to N - 1 not in S:
23                 delta = (M[i][c]*M[r][j] - M[j][c]*M[r][i])
24                 M[r][c] = (M[r][c] - delta * inv) mod p
25     return S // |S| = 2K, Max Matching Size = K
26

```

```

27 // Reconstruct K matched pairs from full-rank submatrix
28 function rabin_vazirani(G, T, S):
29     A = invert_matrix(T[S][S]) // O(|S|^3) Gauss-Jordan
30     matching = []
31     while S has unmatched vertices:
32         pick unmatched u in S
33         find unmatched v in S (adj(u, v) and A[u][v] != 0)
34         matching.append((u, v))
35         // Schur complement update in O(|S|^2)
36         inv = (A[u][v])^(-1) mod p
37         for each unmatched x in S:
38             for each unmatched y in S:
39                 delta = A[x][u]*A[y][v] - A[x][v]*A[y][u]
40                 A[x][y] = (A[x][y] + delta * inv) mod p
41     return matching

```

2.6 Biconnected Component

Complexity: $\mathcal{O}(V + E)$

BCC: Cut vertices, bridges, and 2-connected components

```

1 // Complexity: O(V + E) time
2 // Finds Bridges, Articulation Points, and 2-Connected Components (
  Biconnected Components)
3
4 int num[N], low[N], timer = 0
5 stack<pair<int, int>> edge_stack
6
7 function dfs_bcc(u, p):
8     num[u] = low[u] = ++timer
9     children = 0
10    for v in adj[u]:
11        if v == p: continue
12        if num[v]:
13            low[u] = min(low[u], num[v])
14            if num[v] < num[u]: edge_stack.push({u, v})
15        else:
16            children += 1
17            edge_stack.push({u, v})
18            dfs_bcc(v, u)
19            low[u] = min(low[u], low[v])
20
21            // Bridge condition: low[v] > num[u]
22            if low[v] > num[u]: mark_bridge(u, v)
23
24            // Articulation point condition:
25            if (p == -1 and children > 1) or (p != -1 and low[v] >=
  num[u]):
26                mark_articulation_point(u)
27
28            // Extract Biconnected Component:
29            if low[v] >= num[u]:
30                component = []
31                while True:
32                    edge = edge_stack.pop()
33                    component.append(edge)
34                    if edge == {u, v}: break
35            process_bcc(component)

```

2.7 Functional Cycle

Complexity: $\mathcal{O}(\text{dist} + \text{len})$

Floyd Tortoise and Hare: Find cycle in functional graph $f(x)$

```

1 // Complexity: O(k) time where k is distance + cycle length, O(1)
  space

```

```

2 // Floyd's Tortoise and Hare for functional graph (out-degree of
  each node is 1: f(x))
3
4 function find_cycle(x0, f):
5     tortoise = f(x0), hare = f(f(x0))
6     while tortoise != hare:
7         tortoise = f(tortoise)
8         hare = f(f(hare))
9
10    // Find entry point of cycle (mu)
11    mu = 0, tortoise = x0
12    while tortoise != hare:
13        tortoise = f(tortoise)
14        hare = f(hare)
15        mu += 1
16
17    // Find cycle length (lambda)
18    lam = 1, hare = f(tortoise)
19    while tortoise != hare:
20        hare = f(hare)
21        lam += 1
22
23    return (entry_node = tortoise, start_dist = mu, cycle_len = lam)

```

2.8 Strongly Connected Components

Complexity: $\mathcal{O}(V + E)$

Tarjan SCC on directed graphs

```

1 // Complexity: O(V + E) time, O(V) space
2 // Tarjan's Strongly Connected Components algorithm for directed
  graphs
3
4 int dfn[N], low[N], scc_id[N], timer = 0, scc_cnt = 0
5 stack<int> st; bool in_st[N]
6
7 function dfs_scc(u):
8     dfn[u] = low[u] = ++timer
9     st.push(u); in_st[u] = True
10    for v in adj[u]:
11        if not dfn[v]:
12            dfs_scc(v)
13            low[u] = min(low[u], low[v])
14        else if in_st[v]:
15            low[u] = min(low[u], dfn[v])
16
17    if low[u] == dfn[u]:
18        scc_cnt += 1
19        repeat:
20            v = st.pop(); in_st[v] = False
21            scc_id[v] = scc_cnt
22        until v == u
23
24 function find_all_sccs():
25     fill(dfn, 0); fill(in_st, False)
26     timer = 0; scc_cnt = 0
27     for u = 1 to N:
28         if not dfn[u]: dfs_scc(u)
29     return scc_id

```

2.9 Offline Deletion

Complexity: $\mathcal{O}(T(n) \log n)$

DSU Rollback with segment tree divide & conquer

```

1 // Complexity: O(log N) per union / rollback (No path compression,
  rank by size)
2 // Supports offline dynamic connectivity & edge deletions via
  segment tree divide & conquer
3
4 struct Operation:
5     u, v, united (bool)
6
7 stack<Operation> history
8 int parent[N], sz[N], components_cnt = N
9
10 function find(u):
11     while u != parent[u]: u = parent[u]
12     return u
13
14 function unite(u, v):
15     u = find(u); v = find(v)
16     if u == v:
17         history.push({u, v, False})
18         return False
19     if sz[u] < sz[v]: swap(u, v)
20     parent[v] = u; sz[u] += sz[v]; components_cnt -= 1
21     history.push({u, v, True})
22     return True
23
24 function rollback():
25     op = history.pop()
26     if op.united:
27         sz[op.u] -= sz[op.v]
28         parent[op.v] = op.v
29         components_cnt += 1

```

2.10 2-SAT

Complexity: $\mathcal{O}(n + m)$

Boolean 2-Satisfiability via implication graph and SCC.

- **At least one is true** ($X \vee Y$): Add edges $\neg X \rightarrow Y$ and $\neg Y \rightarrow X$.
- **If X then Y** ($X \implies Y$): Add edges $X \rightarrow Y$ and $\neg Y \rightarrow \neg X$.
- **Exactly one is true** ($X \text{ XOR } Y = 1$): ($X \vee Y$) $\wedge$ ($\neg X \vee \neg Y$). Edges: $X \leftrightarrow \neg Y$ and $\neg X \leftrightarrow Y$.
- **X and Y must be the same** ($X \text{ XOR } Y = 0$): ($X \vee \neg Y$) $\wedge$ ($\neg X \vee Y$). Edges: $X \leftrightarrow Y$ and $\neg X \leftrightarrow \neg Y$.
- **X MUST be true / MUST be false**: Add edge $\neg X \rightarrow X$ (forces true), or $X \rightarrow \neg X$ (forces false).
- **At most one of $\{X_1, X_2, \dots, X_k\}$ is true**: Add prefix variables $P_i = P_{i-1} \vee X_i$. Edges: $X_i \rightarrow P_i$, $P_{i-1} \rightarrow P_i$, and $X_i \rightarrow \neg P_{i-1}$ (if X_i is true, previous must all be false).

```

1 // Complexity: O(N + M) time
2 // 2-SAT: Variable xi is mapped to 2*i (True) and 2*i + 1 (False).
  Negation of u is u ^ 1.
3
4 function add_clause_or(u, v):
5     // (u OR v) <=> (NOT u => v) AND (NOT v => u)
6     adj[u ^ 1].append(v)
7     adj[v ^ 1].append(u)
8
9 function solve_2sat(num_vars):
10    find_all_sccs() // on 2 * num_vars nodes
11    assignment[1..num_vars]
12    for i = 1 to num_vars:
13        if scc_id[2 * i] == scc_id[2 * i + 1]:
14            return (satisfiable = False, {}) // Contradiction
15        // In topological order of DAG of SCCs, later SCC gets
  chosen:
16        assignment[i] = (scc_id[2 * i] < scc_id[2 * i + 1])
17    return (satisfiable = True, assignment)

```

2.11 Floyd

Complexity: $O(n^3)$

All-pair Shortest-path / transitive closure / min-max path

```
1 // Complexity:  $O(V^3)$  time,  $O(V^2)$  space
2 // All-Pairs Shortest Path and Transitive Closure / Min-Max Path
3
4 function floyd_warshall(dist[N][N]):
5     // dist[i][j] initialized to edge weight or INF, dist[i][i] = 0
6     for k = 1 to N:
7         for i = 1 to N:
8             for j = 1 to N:
9                 if dist[i][k] < INF and dist[k][j] < INF:
10                    dist[i][j] = min(dist[i][j], dist[i][k] + dist[
11                        k][j])
12
13     // Negative cycle detection: if dist[i][i] < 0 for any i
```

2.12 Eulerian Path

Complexity: $O(m)$

Hierholzer algorithm: Find Eulerian Path/Circuit.

- **Words/Strings Chaining (De Bruijn / Dominoes):** Chaining words where the end of one matches the start of the next. Model *states* (e.g., prefix/suffix of length $L - 1$, or domino halves) as vertices, and the *items* themselves as directed edges. Eulerian path = sequence covering all items exactly once.
- **Assigning Directions to Undirected Edges:** Goal is to orient edges to balance in-degrees and out-degrees (e.g., $|in - out| \leq 1$). Add dummy edges (or connect odd degree vertices to a dummy node) to make all degrees even, find an Eulerian circuit, and orient edges according to the traversal direction.
- **Edge-Disjoint Cycle Decomposition:** Decompose graph into edge-disjoint cycles. Possible $\iff$ graph is connected and all vertices have even degrees. Finding an Eulerian circuit directly yields the cycles.
- **Bipartite Grid Traversal (Row-Col Graphs):** Matrix problems where you alternate moving along rows and columns. Create a bipartite graph of Rows and Columns. A cell (r, c) is an edge between row r and column c . An Eulerian path traverses cells alternating row-col-row.

```
1 // Complexity:  $O(V + E)$  time and space
2 // Hierholzer's algorithm for finding Eulerian Path / Circuit
3
4 function find_eulerian_path(start_node):
5     path = [], stack st
6     st.push(start_node)
7
8     while not st.empty():
9         u = st.top()
10        if out_degree[u] > 0:
11            // Take an unused edge (u, v), mark edge used
12            v = get_next_unused_edge(u)
13            st.push(v)
14        else:
15            path.append(u)
16            st.pop()
17
18     return reverse(path)
```

2.13 Vizing Theorem

Complexity: $O(n^2)$

Color graph edges with at most $(\Delta + 1)$ colors

```
1 // Complexity:  $O(V * E)$  or  $O(V^2)$  time
2 // Colors the edges of a simple graph using at most  $(\Delta + 1)$ 
   colors (Vizing's theorem)
3
4 function color_edge(u, v0):
5     // Construct fan F = [v0, v1, ..., vk] around u
6     fan = [v0]
7     while can_extend_fan(fan):
8         extend_fan(fan)
9     c = first_color_free_at(u)
10    d = first_color_free_at(last_fan_node)
11    invert_kempe_chain(u, c, d)
12    shift_fan_colors(fan)
13    assign_color(u, v0, d)
14
15 function vizing_edge_coloring():
16    max_deg = max_degree(G)
17    for (u, v) in edges:
18        color_edge(u, v)
19    return edge_colors
```

2.14 Bipartite Online Checking

Complexity: $O(\alpha(n))$

Online bipartite checking upon adding edges via parity DSU

```
1 // Complexity:  $O(\alpha(N))$  per edge addition
2 // Maintain 2-coloring and bipartite validity online using parity-
   weighted DSU
3
4 int parent[N], parity[N], is_bipartite = True
5
6 function find_root(u):
7     if parent[u] == u: return {root: u, par: 0}
8     res = find_root(parent[u])
9     parent[u] = res.root
10    parity[u] = parity[u] ^ res.par
11    return {root: parent[u], par: parity[u]}
12
13 function add_edge(u, v):
14    ru = find_root(u), rv = find_root(v)
15    if ru.root == rv.root:
16        if ru.par == rv.par: // Same color connected by an edge =>
           Odd cycle!
17            is_bipartite = False
18        else:
19            parent[ru.root] = rv.root
20            parity[ru.root] = ru.par ^ rv.par ^ 1
```

2.15 Gomory-Hu Tree

Complexity: $O(n \cdot \text{MaxFlow})$

Equivalent cut tree for all-pairs max flow / min cut

```
1 // Complexity:  $O(V * \text{MaxFlow})$  time
2 // Builds equivalent tree where min capacity on tree path between u
   and v equals Min-Cut in G
3
4 function build_gomory_hu_tree(vertices V):
5     parent[1..N] = 1, weight[1..N] = INF
6     for u = 2 to N:
7         p = parent[u]
8         // Compute s-t min cut between u and p in original graph
9         cut_val, reachable_nodes = dinic_min_cut(u, p)
10        weight[u] = cut_val
11
12    for v = u + 1 to N:
```

```
13        if v in reachable_nodes and parent[v] == p:
14            parent[v] = u
15
16    // Tree edges are (u, parent[u]) with weight[u] for all u in
   [2..N]
```

2.16 Critical (Dominator Tree)

Complexity: $O(m \log n)$

Lengauer-Tarjan: Immediate dominators from source s

```
1 // Complexity:  $O(M \log N)$  or  $O(M * \alpha(N))$  time (Lengauer-Tarjan
   algorithm)
2 // Immediate Dominator idom[u]: vertex d such that every path from
   source s to u must pass through d
3
4 function lengauer_tarjan(source s):
5     dfs_order(s) // compute dfn[u], parent[u], sdом[u]
6     for i = N down to 2:
7         u = vertex_at_dfn[i]
8         for v in pred[u]:
9             sdом[u] = min(sdом[u], sdом[eval(v)])
10        bucket[vertex_at_dfn[sdom[u]]].append(u)
11        link(parent[u], u)
12        for v in bucket[parent[u]]:
13            u_min = eval(v)
14            idom[v] = parent[u] if sdом[u_min] == sdом[v] else
   u_min
15        bucket[parent[u]].clear()
16
17    for i = 2 to N:
18        u = vertex_at_dfn[i]
19        if idom[u] != vertex_at_dfn[sdom[u]]:
20            idom[u] = idom[idom[u]]
```

2.17 Prufer

Complexity: $O(n \log n)$

Encode and decode labeled trees of size n

```
1 // Complexity:  $O(N \log N)$  with priority queue or  $O(N)$  linear time
2 // Prufer Code: Bijective encoding between labeled trees of size N
   and sequences of length N - 2
3
4
5 function encode_prufer(tree, N):
6     deg = [len(adj[i]) for i in 1..N]
7     leaves = min_priority_queue([i for i in 1..N if deg[i] == 1])
8     code = []
9     for step = 1 to N - 2:
10        leaf = leaves.pop()
11        neighbor = get_only_neighbor(leaf)
12        code.append(neighbor)
13        deg[neighbor] -= 1
14        if deg[neighbor] == 1: leaves.push(neighbor)
15    return code
16
17 function decode_prufer(code, N):
18    deg = count_frequencies(code) + 1 for each node
19    leaves = min_priority_queue([i for i in 1..N if deg[i] == 1])
20    edges = []
21    for c in code:
22        leaf = leaves.pop()
23        edges.append({leaf, c})
24        deg[c] -= 1
25        if deg[c] == 1: leaves.push(c)
26    edges.append({leaves.pop(), leaves.pop()})
27    return edges
```

2.18 Steiner Tree

Complexity: $O(n \cdot 3^k + m \cdot 2^k \log m)$
Minimum Steiner Tree connecting k terminal vertices

```
1 // Complexity: O(N * 3^K + M * 2^K log M) time
2 // Minimum weight connected subgraph containing a designated
  terminal set T of size K <= 12
3
4 function steiner_tree(graph, terminals T):
5     K = len(T)
6     dp[1 << K][N] = INF
7     for i = 0 to K - 1:
8         dp[1 << i][T[i]] = 0
9
10    for mask = 1 to (1 << K) - 1:
11        // Transition 1: Split mask into submasks
12        for sub = (mask - 1) & mask; sub > 0; sub = (sub - 1) &
mask:
13            for u = 1 to N:
14                dp[mask][u] = min(dp[mask][u], dp[sub][u] + dp[mask
^ sub][u])
15
16        // Transition 2: Dijkstra across the graph
17        priority_queue pq
18        for u = 1 to N:
19            if dp[mask][u] < INF: pq.push({dp[mask][u], u})
20        while not pq.empty():
21            d, u = pq.pop()
22            if d > dp[mask][u]: continue
23            for (v, w) in adj[u]:
24                if dp[mask][v] > dp[mask][u] + w:
25                    dp[mask][v] = dp[mask][u] + w
26                    pq.push({dp[mask][v], v})
27
28    return min_{u} dp[(1 << K) - 1][u]
```

2.19 Stable Marriage

Complexity: $O(n^2)$
Gale-Shapley Stable Marriage Problem

```
1 // Complexity: O(N^2) time
2 // Gale-Shapley Algorithm: Finds stable matching between N men and
  N women
3
4 function gale_shapley(men_pref, women_pref):
5     husband_of[1..N] = 0, wife_of[1..N] = 0
6     next_proposal_idx[1..N] = 1
7     free_men_queue = [1..N]
8
9     while not free_men_queue.empty():
10        m = free_men_queue.pop()
11        w = men_pref[m][next_proposal_idx[m]++]
12
13        if husband_of[w] == 0:
14            husband_of[w] = m; wife_of[m] = w
15        else:
16            m_current = husband_of[w]
17            if rank[w][m] < rank[w][m_current]: // w prefers m over
m_current
18                husband_of[w] = m; wife_of[m] = w
19                free_men_queue.push(m_current)
20            else:
21                free_men_queue.push(m)
22
23    return (wife_of, husband_of)
```

3 Tree

3.1 Tree Isomorphism

Complexity: $O(n \log n)$
Tree hashing: Check isomorphic rooted trees or unrooted trees (using centroids as roots)

```
1 // Complexity: O(N log N) or O(N) with randomized hash mapping
2 // Assigns a unique hash to rooted trees. For unrooted trees, run
  on tree centroid(s).
3
4 map<vector<int>, int> subtree_hash_map
5
6 function get_tree_hash(u, parent):
7     child_hashes = []
8     for v in adj[u]:
9         if v == parent: continue
10        child_hashes.append(get_tree_hash(v, u))
11    sort(child_hashes)
12    if child_hashes not in subtree_hash_map:
13        subtree_hash_map[child_hashes] = size(subtree_hash_map) + 1
14    return subtree_hash_map[child_hashes]
15
16 // For unrooted tree comparison:
17 function check_isomorphic(T1, T2):
18     c1 = find_centroids(T1), c2 = find_centroids(T2)
19     if len(c1) != len(c2): return False
20     h1 = get_tree_hash(c1[0], -1)
21     for c in c2:
22         if get_tree_hash(c, -1) == h1: return True
23    return False
```

3.2 Centroid Decomposition

Complexity: $O(n \log n)$
Decompose tree into $O(\log n)$ depth hierarchy for divide-and-conquer path queries

```
1 // Complexity: O(N log N) time, O(N) space
2 // Recursively decompose tree at centroid (size of each component
  <= N / 2)
3
4 bool removed[MAXN]; sz[MAXN]
5
6 function get_sizes(u, p):
7     sz[u] = 1
8     for v in adj[u]:
9         if v != p and not removed[v]:
10            sz[u] += get_sizes(v, u)
11    return sz[u]
12
13 function get_centroid(u, p, total_sz):
14     for v in adj[u]:
15         if v != p and not removed[v] and sz[v] > total_sz / 2:
16            return get_centroid(v, u, total_sz)
17    return u
18
19 function decompose(entry_u, parent_centroid):
20     total_sz = get_sizes(entry_u, -1)
21     c = get_centroid(entry_u, -1, total_sz)
22     removed[c] = True
23     centroid_parent[c] = parent_centroid
24
25     // Process paths passing through centroid c
26     solve_for_centroid(c)
27
28    for v in adj[c]:
```

```
29         if not removed[v]:
30             decompose(v, c)
```

3.3 Heavy-Light Decomposition

Complexity: $O(n \log^2 n)$

Decompose tree into heavy chains for $O(\log^2 n)$ path queries/updates and $O(\log n)$ subtree queries

```
1 // Complexity: O(N) preprocess, O(log^2 N) path query (O(log N)
  with Segment Tree)
2 // Decompose tree into heavy chains where each vertex jumps at most
  O(log N) chains to root
3
4 int parent[N], depth[N], sz[N], heavy[N], head[N], pos[N], cur_pos
  = 0
5
6 function dfs_sz(u, p, d):
7     sz[u] = 1, parent[u] = p, depth[u] = d, heavy[u] = -1,
max_child_sz = 0
8     for v in adj[u]:
9         if v == p: continue
10        dfs_sz(v, u, d + 1)
11        sz[u] += sz[v]
12        if sz[v] > max_child_sz: max_child_sz = sz[v], heavy[u] = v
13
14 function dfs_hld(u, h):
15     head[u] = h, pos[u] = ++cur_pos
16     if heavy[u] != -1: dfs_hld(heavy[u], h)
17     for v in adj[u]:
18         if v != parent[u] and v != heavy[u]: dfs_hld(v, v)
19
20 function query_path(u, v):
21     res = identity
22     while head[u] != head[v]:
23         if depth[head[u]] > depth[head[v]]: swap(u, v)
24         res = combine(res, segtree_query(pos[head[v]], pos[v]))
25         v = parent[head[v]]
26     if depth[u] > depth[v]: swap(u, v)
27     res = combine(res, segtree_query(pos[u], pos[v]))
28    return res
```

4 String

4.1 String Problem Solving Strategies

String problem taxonomy, periodicity, LCP, distinct substrings, palindromes & key theorems

• Problem Taxonomy & Core Approches:

- Longest Common Prefix with Whole String / Single Pattern Match: Use Z-Algorithm or KMP.
- Multiple Dictionary Matching / DP on String Sets: Use Aho-Corasick Automaton.
- Distinct Substrings: Suffix Array ($\frac{n(n+1)}{2} - \sum LCP_i$) or SAM ($\sum len[u] - len[link[u]]$).
- k -th Substring / LCS of Multiple Strings: Use Suffix Automaton (SAM) or Suffix Array + LCP.
- Palindromic Substrings: Use Manacher (radius centered at every index) or EERTREE / PAM (palindromic tree).
- Lexicographically Minimal String Rotation: Run Lyndon Factorization (Duval) on $S + S$.
- Fast Substring Equivalence $O(1)$ / 2D Strings: Use Rolling Hash ($2^{61} - 1$ Mersenne prime).

• Periodicity & Border Properties:

- Minimal period of length- n string is $p = n - \pi[n - 1]$ (if $n \bmod p == 0 \implies S = (S[0..p-1])^{n/p}$).
- Fine-Wilf (Periodicity Lemma):** If S has periods p, q and $|S| \geq p + q - \gcd(p, q) \implies S$ has period $\gcd(p, q)$.
- Border Series:** All borders of a string partition into $\leq O(\log n)$ arithmetic progressions $\implies$ Optimize border DP to $\tilde{O}(n \log n)$.

• Theoretical Bounds:

- Distinct palindromic substrings in length- n string is always $\leq n$.
- Suffix Automaton of length- n string has $\leq 2n - 1$ states and $\leq 3n - 4$ transition edges.

4.2 Rolling Hash (61-bit Mersenne)

Complexity: $O(n)$ build, $O(1)$ substring query

Double/61-bit Mersenne prime Hash ($2^{61} - 1$) with randomized base

```
1 // Complexity: O(N) prep, O(1) substring hash query
2 // Modulo: Mersenne prime 2^61 - 1 (fast bitwise modulo, immune to
  hacking)
3
4 typedef unsigned __int128 uint128;
5 const uint64_t MOD = (1ULL << 61) - 1;
6 uint64_t BASE; // Pick random odd in [256, 10^9] using mt19937_64
7
8 function mul_mod(uint64_t a, uint64_t b):
9     uint128 p = (uint128)a * b;
10    uint64_t res = (p >> 61) + (p & MOD);
11    if (res >= MOD) res -= MOD;
12    return res;
13
14 struct StringHash:
15     vector<uint64_t> H, P; // Prefix hashes and powers of BASE
16
17     function init(string s):
18         n = len(s); H.resize(n + 1, 0); P.resize(n + 1, 1);
19         for i = 0 to n - 1:
20             P[i + 1] = mul_mod(P[i], BASE);
21             H[i + 1] = (mul_mod(H[i], BASE) + s[i]) % MOD;
22
23     // Query hash of substring s[l..r] (0-indexed)
24     function query(l, r):
25         uint64_t sub = mul_mod(H[l], P[r - l + 1]);
26         return (H[r + 1] + MOD - sub) % MOD;
```

4.3 KMP

Complexity: $O(n + m)$

Prefix function π and linear string matching

```
1 // Complexity: O(|P|) preprocess, O(|T|) search
2 // pi[i] = length of longest proper prefix of P[0..i] that is also
  a suffix of P[0..i]
3
4 function compute_pi(pattern P):
5     m = length(P), pi[0] = 0, j = 0
6     for i = 1 to m - 1:
7         while j > 0 and P[i] != P[j]:
8             j = pi[j - 1]
9         if P[i] == P[j]:
10            j = j + 1
11            pi[i] = j
12    return pi
13
14 function kmp_search(text T, pattern P):
```

```
15    pi = compute_pi(P)
16    n = length(T), m = length(P), j = 0
17    for i = 0 to n - 1:
18        while j > 0 and T[i] != P[j]:
19            j = pi[j - 1]
20        if T[i] == P[j]:
21            j = j + 1
22        if j == m:
23            report_match_at(i - m + 1)
24            j = pi[j - 1]
```

4.4 Z-Algorithm

Complexity: $O(n + m)$

Z-array: $z[i] = \text{LCP}(S, S[i..n-1])$ and pattern matching via $S = P + \# + T$

```
1 // Complexity: O(N) time and space
2 // Computes Z-array: z[i] = LCP(S, S[i..N-1])
3 // Recognition: Longest common prefix with entire string, Pattern
  matching via S = P + '#' + T
4
5 function compute_z(string s):
6     n = len(s), z = array of size n (all 0)
7     z[0] = n
8     l = 0, r = 0
9     for i = 1 to n - 1:
10        if i <= r:
11            z[i] = min(r - i + 1, z[i - l])
12        while i + z[i] < n and s[z[i]] == s[i + z[i]]:
13            z[i] += 1
14        if i + z[i] - 1 > r:
15            l = i; r = i + z[i] - 1
16    return z
```

4.5 Aho Corasick

Complexity: $O(\sum |P_i| + |T|)$

Multi-pattern dictionary matching on Trie with BFS failure/exit links

```
1 // Complexity: O(\sum |P_i| * |\Signal|) build, O(|T|) search
2 // Multi-pattern string matching on Trie with failure & dictionary
  suffix links
3
4 struct Node:
5     next[ALPHABET], link = 0, exit_link = 0, match_ids = []
6
7 function build_aho_corasick(patterns):
8     root = 0, new_node(root)
9     for id, p in enumerate(patterns):
10        u = root
11        for c in p:
12            if not tree[u].next[c]: tree[u].next[c] = new_node()
13            u = tree[u].next[c]
14        tree[u].match_ids.append(id)
15
16    // BFS to build failure links
17    queue Q
18    for c in ALPHABET:
19        v = tree[root].next[c]
20        if v: Q.push(v); tree[v].link = root
21    while not Q.empty():
22        u = Q.pop()
23        for c in ALPHABET:
24            v = tree[u].next[c]
25            if v:
```

```
26        f = tree[u].link
27        while f != root and not tree[f].next[c]: f = tree[f]
28        ].link
29        tree[v].link = tree[f].next[c] if tree[f].next[c]
30        and tree[f].next[c] != v else root
31        tree[v].exit_link = tree[v].link if tree[tree[v].
32        link].match_ids else tree[tree[v].link].exit_link
33        Q.push(v)
34    else:
35        tree[u].next[c] = tree[tree[u].link].next[c] //
36        Trie transition shortcut
```

4.6 Suffix Array

Complexity: $O(n \log n)$ construction, $O(n)$ LCP
Suffix Array doubling construction and Kasai LCP algorithm

```
1 // Complexity: O(N log N) or O(N log^2 N) for SA, O(N) for Kasai
  LCP
2 // SA[i] = start index of i-th lexicographically smallest suffix;
  LCP[i] = lcp(S[SA[i]..], S[SA[i-1]..])
3
4 function build_suffix_array(string S):
5     S = S + "$" // sentinel smallest char
6     n = length(S)
7     SA[0..n-1] = [0..n-1], rank[0..n-1] = ASCII of S
8
9     k = 1
10    while k < n:
11        // Sort SA by key: (rank[i], rank[(i + k) % n])
12        sort(SA, key = (i) -> (rank[i], rank[(i + k) % n]))
13        tmp_rank[SA[0]] = 0
14        for i = 1 to n - 1:
15            prev = SA[i-1], cur = SA[i]
16            same = (rank[cur] == rank[prev] and rank[(cur + k) % n]
17            == rank[(prev + k) % n])
18            tmp_rank[cur] = tmp_rank[prev] + (0 if same else 1)
19            rank = tmp_rank
20            if rank[SA[n-1]] == n - 1: break
21            k = k * 2
22    return SA
23
24 function kasai_lcp(string S, SA):
25     n = length(S), pos[0..n-1], lcp[0..n-1]
26     for i = 0 to n - 1: pos[SA[i]] = i
27     k = 0
28     for i = 0 to n - 1:
29         if pos[i] == 0: continue
30         j = SA[pos[i] - 1]
31         while i + k < n and j + k < n and S[i + k] == S[j + k]: k =
32         k + 1
33         lcp[pos[i]] = k
34         if k > 0: k = k - 1
35    return lcp
```

• LCP Applications & Core CP Recipes:

- Arbitrary Suffix LCP:** $\text{LCP}(S[i..], S[j..]) = \min_{k=\max(pos[i], pos[j])}^{\min(pos[i], pos[j])+1} \text{LCP}[k]$ via RMQ / Sparse Table in $O(1)$ query.
- Distinct Substrings:** Total distinct substrings is $\frac{n(n+1)}{2} - \sum_{i=1}^{n-1} \text{LCP}[i]$.
- k -th Lexicographical Substring:** Suffix $SA[i]$ contributes $n - SA[i] - \text{LCP}[i]$ new strings of lengths $\text{LCP}[i] + 1 \dots n - SA[i]$.
- LCS of 2 Strings:** Build SA on $S_1 + \# + S_2$; ans is $\max \text{LCP}[i]$ where $SA[i], SA[i-1]$ belong to different strings.

- **LCS of K Strings:** Build SA on $S_1 + \#_1 + \dots + S_K + \#_K$; two pointers / sliding window $[L, R]$ covering all K strings $\Rightarrow \min_{k=L+1}^R LCP[k]$.
- **Pattern Occurrences Range:** Suffixes with prefix P form continuous range $[L, R]$ in SA; find via 2 binary searches in $O(|P| \log n)$, count = $R - L + 1$.
- **Longest Substring Occurring $\geq K$ Times:** $\max_i (\min_{j=i+1}^{i+K-1} LCP[j])$ via monotonic queue / sliding RMQ in $O(n)$.

4.7 Suffix Automaton

Complexity: $O(n|\Sigma|)$

Minimal DAWG where every path from root corresponds to a substring of S

```
1 // Complexity: O(N * |\Sigma|) time, O(N * |\Sigma|) space
2 // Directed Acyclic Word Graph (DAWG) representing all substrings
  of S
3
4 struct State:
5     len = 0, link = -1, next[ALPHABET] = {0}
6
7 State st[2 * MAXN]; sz = 1, last = 0
8
9 function sam_init():
10     st[0].len = 0, st[0].link = -1; sz = 1, last = 0
11
12 function sam_extend(char c):
13     cur = sz++; st[cur].len = st[last].len + 1
14     p = last
15     while p != -1 and not st[p].next[c]:
16         st[p].next[c] = cur
17         p = st[p].link
18
19     if p == -1:
20         st[cur].link = 0
21     else:
22         q = st[p].next[c]
23         if st[p].len + 1 == st[q].len:
24             st[cur].link = q
25         else:
26             clone = sz++
27             st[clone].len = st[p].len + 1
28             st[clone].next = st[q].next
29             st[clone].link = st[q].link
30             while p != -1 and st[p].next[c] == q:
31                 st[p].next[c] = clone
32                 p = st[p].link
33             st[q].link = st[cur].link = clone
34     last = cur
```

4.8 Lyndon Factorization

Complexity: $O(n)$ time, $O(1)$ space

Duval algorithm: factorize string into Lyndon words $s_1 \geq s_2 \geq \dots \geq s_k$ & Minimal string rotation

```
1 // Complexity: O(N) time, O(1) space
2 // Duval's algorithm: factorizes S into Lyndon words s1 >= s2 >=
  ... >= sk (each si is strictly smaller than all its proper
  suffixes)
3
4 function duval_lyndon_factorization(string S):
5     n = length(S), i = 0, factors = []
6     while i < n:
```

```
7         j = i + 1, k = i
8         while j < n and S[k] <= S[j]:
9             if S[k] < S[j]: k = i
10            else: k = k + 1 // periodic continuation
11            j = j + 1
12            while i <= k:
13                factors.append(S[i .. i + j - k - 1])
14            i = i + j - k
15     return factors
```

4.9 Manacher

Complexity: $O(n)$

Find the longest palindrome radius centered at every index

```
1 // Complexity: O(N) time, O(N) space
2 // Finds longest palindrome radius P[i] centered at i for
  transformed string T = "~#a#b#a#$"
3
4 function manacher(string S):
5     // Insert '#' between characters and '~', '$' as boundary
      markers
6     T = "~#"
7     for c in S: T += "#" + c
8     T += "#$"
9
10    n = length(T), P[0..n-1] = 0
11    C = 0, R = 0 // Center and Right boundary of rightmost
      palindrome
12
13    for i = 1 to n - 2:
14        i_mirror = 2 * C - i
15        if R > i:
16            P[i] = min(R - i, P[i_mirror])
17            // Expand palindrome around i
18            while T[i + 1 + P[i]] == T[i - 1 - P[i]]:
19                P[i] = P[i] + 1
20            // Update center and right boundary
21            if i + P[i] > R:
22                C = i
23                R = i + P[i]
24
25    // Palindrome of max length centered at S[idx]: length is P[i],
      start in S is (i - P[i]) / 2
26    return P
```

4.10 Palindrome Tree

Complexity: $O(n)$

EERTREE: Represent all distinct palindromes in string and transition links

```
1 // Complexity: O(N) time and space
2 // EERTREE stores all distinct palindromic substrings. Root 1: len
  -1, Root 2: len 0.
3
4 struct Node:
5     len, link, next[ALPHABET]
6
7 Node tree[MAXN]; sz, last, S = ""
8
9 function eertree_init():
10    tree[1].len = -1; tree[1].link = 1 // Imaginary root for odd
      length
11    tree[2].len = 0; tree[2].link = 1 // Real root for even length
12    sz = 2, last = 2, S = ""
13
```

```
14 function get_link(p):
15     n = length(S)
16     while n - tree[p].len - 2 < 0 or S[n - tree[p].len - 2] != S[n
      - 1]:
17         p = tree[p].link
18     return p
19
20 function eertree_add_char(char c):
21     S += c
22     cur = get_link(last)
23     if not tree[cur].next[c]:
24         now = ++sz
25         tree[now].len = tree[cur].len + 2
26         tree[now].link = 2 if tree[cur].len == -1 else tree[
      get_link(tree[cur].link)].next[c]
27         tree[cur].next[c] = now
28     last = tree[cur].next[c]
```

4.11 De Bruijn

Complexity: $O(k^n)$

Generate cyclic sequence of length k^n containing all k -ary words of length n

```
1 // Complexity: O(K^N) time and space
2 // Constructs cyclic string of length K^N containing all substrings
  of length N over alphabet [0..K-1]
3
4 function de_bruijn(K, N):
5     if K == 1: return "0" * N
6     visited_edges = set()
7     sequence = []
8
9     function dfs(node_str):
10        for c in 0 to K - 1:
11            edge = node_str + str(c)
12            if edge not in visited_edges:
13                visited_edges.insert(edge)
14                dfs(edge[1..N-1])
15                sequence.append(c)
16
17    start_node = "0" * (N - 1)
18    dfs(start_node)
19    // Repeat prefix of length N - 1 at the end to make it cyclic
20    return reverse(sequence) + start_node
```

5 Dynamic Programming

5.1 Reroot DP

Complexity: $O(n)$ for DFS traversal

Template for Rerooting on Tree (2-pass DFS with Prefix/Suffix transitions)

```
1 // Complexity: O(N) time, O(N) space
2 // Compute subtree values first (DFS1), then reroot to compute all
  roots (DFS2)
3
4 Data dp_down[N], dp_all[N]
5
6 function dfs_down(u, parent):
7     dp_down[u] = base_state(u)
8     for v in adj[u]:
9         if v == parent: continue
10        dfs_down(v, u)
11        dp_down[u] = merge(dp_down[u], transition(dp_down[v], u, v))
```

```

12
13 function dfs_reroot(u, parent, from_parent_val):
14     // Prefix & Suffix merge of children transitions for O(1)
15     // exclusion
16     children = [v for v in adj[u] if v != parent]
17     deg = len(children)
18     pref[0..deg+1] = identity, suff[0..deg+1] = identity
19
20     for i = 1 to deg:
21         pref[i] = merge(pref[i-1], transition(dp_down[children[i-1]], u, children[i-1]))
22     for i = deg down to 1:
23         suff[i] = merge(suff[i+1], transition(dp_down[children[i-1]], u, children[i-1]))
24
25     dp_all[u] = merge(pref[deg], from_parent_val)
26
27     for i = 1 to deg:
28         v = children[i-1]
29         without_v = merge(pref[i-1], suff[i+1])
30         up_to_v = merge(without_v, from_parent_val)
31         dfs_reroot(v, u, transition(up_to_v, v, u))

```

5.2 Divide-And-Conquer DP & Quadrangle Inequality

Complexity: $\mathcal{O}(kn \log n)$ for DnC / $\mathcal{O}(n^2)$ for Knuth Quadrangle Inequality (Monge): $\forall a \leq b \leq c \leq d \implies C(a, c) + C(b, d) \leq C(a, d) + C(b, c)$ (Cross $\leq$ Nest).

- **Sufficient & Necessary Local Test:** Checking adjacent 2×2 cells is sufficient (telescoping sum):

$$C(i, j+1) - C(i, j) \geq C(i+1, j+1) - C(i+1, j) \quad (\forall i < j)$$

(Equivalently, cost increase $\Delta_j(i) = C(i, j+1) - C(i, j)$ is non-increasing in i ; or continuous $\frac{\partial^2 C}{\partial x \partial y} \leq 0$).

- **Common Cost Functions Satisfying QI:**

- $C(l, r) = f(S_r - S_{l-1})$ for any convex f ($f'' \geq 0$), e.g. $(S_r - S_{l-1})^2$, $(S_r - S_{l-1})^k$ ($k \geq 1$).
- $C(l, r) = \sum_{k=l}^r w_k = W_r - W_{l-1}$ ($w_k \geq 0$, exact equality = in Knuth Optimal BST).
- $C(l, r) = \text{inv}(l, r)$ (inversion count in subarray $[l, r]$, or any cross-pair property).
- $C(l, r) = |x_r - x_l|^p$ ($p \geq 1$ on sorted x), or variance / sum of squared differences.

- **Monotonicity:** Monge property $\implies \text{opt}(i_1) \leq \text{opt}(i_2)$ for $i_1 \leq i_2$ (DnC: $\mathcal{O}(kn \log n)$) and $\text{opt}[i][j-1] \leq \text{opt}[i][j] \leq \text{opt}[i+1][j]$ (Knuth: $\mathcal{O}(n^2)$).

```

1 // Complexity: O(K * N log N) for DnC, O(N^2) for Knuth
2 // Optimization
3 // Condition: opt[i][j-1] <= opt[i][j] <= opt[i+1][j] (Quadrangle
4 // Inequality)
5 // Recurrence: dp[k][i] = min_{j < i} (dp[k-1][j] + cost(j+1, i))
6
7 function compute(k, L, R, optL, optR):
8     if L > R: return
9     mid = (L + R) / 2
10    best_opt = -1
11    dp[k][mid] = INFINITY
12
13    // Search best transition j in [optL, min(mid-1, optR)]
14    for j = optL to min(mid - 1, optR):

```

```

13    val = dp[k-1][j] + cost(j + 1, mid)
14    if val < dp[k][mid]:
15        dp[k][mid] = val
16        best_opt = j
17
18    // Divide and Conquer to subproblems
19    compute(k, L, mid - 1, optL, best_opt)
20    compute(k, mid + 1, R, best_opt, optR)
21
22    // Knuth's Optimization (O(N^2)):
23    // For len = 2 to N:
24    //     For i = 1 to N - len + 1, j = i + len - 1:
25    //         dp[i][j] = min_{k in [opt[i][j-1], opt[i+1][j]]} (dp[i][k] +
26    //             dp[k+1][j] + cost(i, j))

```

5.3 1D/1D DP Optimization

Complexity: $\mathcal{O}(n \log n)$
Optimize $\text{dp}[i] = \min_{j < i} (\text{dp}[j] + \text{cost}(j, i))$ satisfying quadrangle inequality via Deque of optimal intervals

```

1 // Complexity: O(N log N) time
2 // Recognition: 1D DP dp[i] = min_{j < i} (dp[j] + cost(j, i))
3 // satisfying Quadrangle Inequality
4
5 struct Interval:
6     j, l, r // Candidate j is optimal for all indices in [l, r]
7
8 function optimize_1did(int N):
9     deque<Interval> dq
10    dq.push_back(Interval(0, 1, N))
11    dp[0] = 0
12
13    function eval(j, i): return dp[j] + cost(j, i)
14
15    for i = 1 to N:
16        while not dq.empty() and dq.front().r < i:
17            dq.pop_front()
18        dq.front().l = i
19        dp[i] = eval(dq.front().j, i)
20        if i == N: break
21
22        // Binary search where candidate i beats existing
23        // candidates from back
24        while not dq.empty() and eval(i, dq.back().l) <= eval(dq.back().j, dq.back().l):
25            dq.pop_back()
26
27        if dq.empty():
28            dq.push_back(Interval(i, i + 1, N))
29        else:
30            // Binary search intersection point in dq.back()
31            low = dq.back().l, high = dq.back().r, split = high + 1
32            while low <= high:
33                mid = (low + high) / 2
34                if eval(i, mid) <= eval(dq.back().j, mid):
35                    split = mid; high = mid - 1
36                else: low = mid + 1
37            dq.back().r = split - 1
38            if split <= N:
39                dq.push_back(Interval(i, split, N))
40
41    return dp[N]

```

5.4 Alien DP

Complexity: $\mathcal{O}(n \log(\max_val))$

Dynamic Programming on convex/concave functions (WQS Binary Search):

- Cost is convex/concave $\rightarrow$ Greedy selection.
- DP is convex/concave $\rightarrow$ Binary search penalty λ .

```

1 // Complexity: O(N log(C))
2 // Problem: Optimize f(K) = min cost choosing exactly K items,
3 // where f is convex/concave.
4 // Trick: Introduce penalty lambda for each item chosen: g(lambda)
5 // = min_x (cost(x) - lambda * cnt(x))
6
7 function solve_with_penalty(lambda):
8     // dp[i] stores pair (optimal_cost, count_of_items_chosen)
9     dp[0] = (0, 0)
10    for i = 1 to N:
11        dp[i] = dp[i-1] // case: don't choose item i
12        take_cost = dp[i-1].cost + value[i] - lambda
13        take_cnt = dp[i-1].cnt + 1
14        if take_cost < dp[i].cost: // or <= depending on tie-breaking
15            dp[i] = (take_cost, take_cnt)
16    return dp[N]
17
18 function aliens_trick(target_K):
19    low = -MAX_VAL, high = MAX_VAL, best_lambda = 0
20    while low <= high:
21        mid = (low + high) / 2
22        res = solve_with_penalty(mid)
23        if res.cnt >= target_K:
24            best_lambda = mid
25            high = mid - 1 // or low = mid + 1 for max problem
26        else:
27            low = mid + 1
28    ans = solve_with_penalty(best_lambda)
29    return ans.cost + target_K * best_lambda

```

5.5 Convex Hull Trick

Complexity: $\mathcal{O}(n \log n)$ or $\mathcal{O}(n)$ with monotonic slopes
Dynamic Programming optimization for linear functions $y = mx + c$

```

1 // Complexity: O(N log N) or O(N) if slopes are monotonic
2 // Form: dp[i] = min_{j < i} (m[j] * x[i] + c[j])
3
4 struct Line:
5     m, c // y = m * x + c
6
7 // Intersection x-coordinate between line 1 and line 2
8 function intersect_x(l1, l2):
9     return (l2.c - l1.c) / (l1.m - l2.m)
10
11 // Monotonic Slope Deque / Stack (O(N)):
12 deque<Line> hull
13
14 function add_line(new_line):
15     // If slopes decreasing and finding min:
16     while size(hull) >= 2 and intersect_x(hull[last], new_line) <=
17         intersect_x(hull[last-1], hull[last]):
18         hull.pop_back()
19     hull.push_back(new_line)
20
21 function query_min(x):
22     // Two pointers (if query x is increasing):
23     while size(hull) >= 2 and hull[0].eval(x) >= hull[1].eval(x):
24         hull.pop_front()
25     return hull[0].eval(x)
26
27 // Or binary search on hull if x is arbitrary (O(log N))

```

5.6 Slope Trick

Complexity: $\mathcal{O}(n \log n)$

Maintain continuous piecewise linear convex function $f(x)$ with $+|x - a|$, prefix min, domain shift via 2 Heaps

```
1 // Complexity:  $\mathcal{O}(N \log N)$  time
2 // Recognition: DP with convex piecewise linear function,  $f(x) += |x - a|$ , prefix min
3
4 struct SlopeTrick:
5     max_pq L; min_pq R // Multiset of inflection points
6     min_f = 0, l_add = 0, r_add = 0
7
8     // Add  $|x - a|$  to function  $f(x)$ 
9     function add_abs(a):
10         l_max = L.empty() ? -INF : L.top() + l_add
11         r_min = R.empty() ? INF : R.top() + r_add
12         if a < l_max:
13             min_f += l_max - a
14             L.pop(); L.push(a - l_add); L.push(a - l_add)
15             R.push(l_max - r_add)
16         else if a > r_min:
17             min_f += a - r_min
18             R.pop(); R.push(a - r_add); R.push(a - r_add)
19             L.push(r_min - l_add)
20         else:
21             L.push(a - l_add); R.push(a - r_add)
22
23     // Prefix minimum:  $g(x) = \min_y \{x \leq y\} f(y)$  (clears right slopes)
24     function prefix_min():
25         R.clear()
26
27     // Shift function domain  $[x - b, x + a]$ :  $g(x) = \min_{\{x - b \leq y \leq x + a\}} f(y)$ 
28     function shift(a, b):
29         l_add -= b; r_add += a
```

5.7 SOS DP

Complexity: $\mathcal{O}(N \cdot 2^N)$

Sum Over Subsets / Fast Zeta Transform: Compute sum over submasks / supermasks in $\mathcal{O}(N \cdot 2^N)$ ($N \leq 24$)

```
1 // Complexity:  $\mathcal{O}(N \cdot 2^N)$  time,  $\mathcal{O}(2^N)$  space ( $N \leq 24$ )
2 // Recognition: Sum over subsets/supersets, counting pairs with  $[A[i] \text{ AND } A[j]] == 0$ 
3
4 function sos_dp(int N, array F):
5     // F[mask] initialized with base values
6     // Sum over all submasks:  $F[\text{mask}] = \sum_{\text{sub} \subseteq \text{mask}} A[\text{sub}]$ 
7     for i = 0 to N - 1:
8         for mask = 0 to  $(1 \ll N) - 1$ :
9             if mask & (1 << i):
10                 F[mask] += F[mask ^ (1 << i)]
11
12     // Sum over all supermasks:  $G[\text{mask}] = \sum_{\text{mask} \subseteq \text{sup}} A[\text{sup}]$ 
13     // for i = 0 to N - 1:
14     //     for mask =  $(1 \ll N) - 1$  down to 0:
15     //         if not (mask & (1 << i)):
16     //             G[mask] += G[mask | (1 << i)]
17     return F
```

5.8 Monotonic Queue Optimization

Complexity: $\mathcal{O}(1)$ amortized per push/pop

Deque / Two-Stack Sliding Window minimum/maximum

```
1 // Complexity:  $\mathcal{O}(1)$  amortized per push/pop/get_min
2 // Two-stack Min Queue or Monotonic Deque for Sliding Window
3
4 struct MinQueue:
5     deque<pair<value, index>> dq
6
7     function push(val, idx):
8         while not dq.empty() and dq.back().value >= val:
9             dq.pop_back()
10        dq.push_back({val, idx})
11
12    function pop(min_valid_idx):
13        while not dq.empty() and dq.front().index < min_valid_idx:
14            dq.pop_front()
15
16    function get_min():
17        return dq.front().value
```

5.9 Expected Value Trick

Complexity: $\mathcal{O}(n\sqrt{n})$

Suppose we optimize picking k objects from n randomly shuffled items: at step i , only consider state range $\left[\frac{i-k}{n} - C\sqrt{n}, \frac{i-k}{n} + C\sqrt{n}\right]$.

```
1 // Complexity:  $\mathcal{O}(N \cdot \sqrt{N})$ 
2 // Problem: Pick K items from N shuffled items optimizing cost.
3 // Key Insight: The number of items chosen after prefix i concentrates around  $E[X] = (i * K) / N$ .
4 // With high probability, optimal count at step i lies in  $[(i*K)/N - C*\sqrt{N}, (i*K)/N + C*\sqrt{N}]$ .
5
6 function solve_ev_dp(N, K):
7     dp[0..1][0..K] = -INFINITY
8     dp[0][0] = 0
9
10    for i = 1 to N:
11        cur = i % 2, prev = 1 - cur
12        fill(dp[cur], -INFINITY)
13
14        expected_cnt = (i * K) / N
15        low_cnt = max(0, expected_cnt - C * sqrt(N))
16        high_cnt = min(min(i, K), expected_cnt + C * sqrt(N))
17
18        for j = low_cnt to high_cnt:
19            // Transition 1: don't take item i
20            dp[cur][j] = max(dp[cur][j], dp[prev][j])
21            // Transition 2: take item i
22            if j > 0 and dp[prev][j-1] != -INFINITY:
23                dp[cur][j] = max(dp[cur][j], dp[prev][j-1] + val[i])
24
25    return dp[N % 2][K]
```

6 Math

6.1 Number Theory Formulas

Modular arithmetic, Extended Euler Power Reduction, Legendre, Wilson, Frobenius Coin, Floor Trick, CRT, Fibonacci

- Extended Euler Totient Power Reduction ($\forall a, m$):

$$a^b \equiv a^{(b \bmod \phi(m)) + \phi(m)} \pmod{m} \quad (b \geq \phi(m))$$

- Legendre's Formula ($v_p(n!)$): $v_p(n!) = \sum_{k=1}^{\infty} \lfloor \frac{n}{p^k} \rfloor = \frac{n - S_p(n)}{p-1}$.
- Wilson's Theorem: $(p-1)! \equiv -1 \pmod{p} \iff p$ prime $\implies (p-2)! \equiv 1 \pmod{p}$.
- Frobenius Coin (Chicken McNugget): For $\gcd(a, b) = 1$, max non-representable $ax+by$ ($x, y \geq 0$) is $ab-a-b$; total non-representable count is $\frac{(a-1)(b-1)}{2}$.
- Floor Trick ($\mathcal{O}(\sqrt{n})$ Blocks):

```
1 for (int l = 1, r; l <= n; l = r + 1) {
2     int v = n / l; r = n / v;
3 }
```

- Chinese Remainder Theorem (CRT): $x \equiv a_i \pmod{m_i}$.

$$x \equiv \sum_{i=1}^k a_i M_i (M_i^{-1} \bmod m_i) \pmod{M} \quad (M = \prod m_i, M_i = M/m_i)$$

- Fibonacci: $\gcd(F_m, F_n) = F_{\gcd(m, n)}$, $F_{2n} = F_n(2F_{n+1} - F_n)$.

6.2 Like-Euclid Floor Sum

Complexity: $\mathcal{O}(\log m)$

Compute $\sum_{i=0}^{n-1} \lfloor \frac{ai+b}{m} \rfloor$ via coordinate transformation

```
1 // Complexity:  $\mathcal{O}(\log m)$ 
2 // Computes  $\sum_{i=0}^{n-1} \lfloor \frac{a * i + b}{m} \rfloor$  (Like-Euclid)
3 // Recognition: Summing floor divisions of linear functions over ranges
4
5 function floor_sum(n, m, a, b):
6     ans = 0
7     if a >= m:
8         ans += (n - 1) * n / 2 * (a / m)
9         a %= m
10    if b >= m:
11        ans += n * (b / m)
12        b %= m
13    y_max = (a * n + b) / m
14    x_max = y_max * m - b
15    if y_max == 0: return ans
16    // Rotate coordinate system: sum over y from 0 to y_max - 1
17    ans += (n - (x_max + a - 1) / a) * y_max
18    ans += floor_sum(y_max, a, m, (a - x_max % a) % a)
19    return ans
```

6.3 Rabin-Miller

Complexity: $\mathcal{O}(k \log n)$

Deterministic Miller-Rabin prime checking for $n < 2^{64}$

```
1 // Complexity:  $\mathcal{O}(k \log^3 n)$  or  $\mathcal{O}(k \log n)$  with fast multiplication
2 // Deterministic for  $n < 2^{64}$  using bases: {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37}
3
4 function power_mod(a, b, m):
5     res = 1; a %= m
6     while b > 0:
7         if b & 1: res = (res * a) % m
```

```

8     a = (a * a) % m; b >= 1
9     return res
10
11 function miller_test(d, n, a):
12     x = power_mod(a, d, n)
13     if x == 1 or x == n - 1: return True
14     while d != n - 1:
15         x = (x * x) % n; d *= 2
16         if x == n - 1: return True
17         if x == 1: return False
18     return False
19
20 function is_prime(n):
21     if n < 2: return False
22     if n <= 3: return True
23     if n % 2 == 0: return False
24     d = n - 1
25     while d % 2 == 0: d /= 2
26     for a in [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37]:
27         if n == a: return True
28         if not miller_test(d, n, a): return False
29     return True

```

6.4 Fast Sieve

Complexity: $O(N)$

Linear Euler Sieve computing primes and smallest prime factors up to N

```

1 // Complexity:  $O(N)$  time and  $O(N)$  space (Linear Euler Sieve)
2 // Finds all primes up to  $N$  and smallest prime factor min_prime[i]
3
4 int min_prime[N], primes[]
5
6 function linear_sieve(N):
7     for i = 2 to N:
8         if min_prime[i] == 0:
9             min_prime[i] = i
10            primes.append(i)
11        for p in primes:
12            if p > min_prime[i] or i * p > N: break
13            min_prime[i * p] = p
14    return primes

```

6.5 Prime Counting

Complexity: $O(N^{2/3})$ or $O(N^{3/4})$

Lucy Hedgehog / Meissel-Lehmer: Count primes $\leq N$

```

1 // Complexity:  $O(N^{3/4})$  or  $O(N^{2/3})$  with Lucy Hedgehog
  algorithm
2 // Counts number of primes  $\leq N$  without generating all primes
3
4 function count_primes(N):
5     sq = sqrt(N)
6     V = [N / i for i in 1..N] // All distinct values of floor(N / i)
7     // S[v] stores number of integers  $\leq v$  with prime factors  $\geq p$ 
8     S = map()
9     for v in V: S[v] = v - 1
10
11    for p = 2 to sq:
12        if S[p] > S[p - 1]: // p is prime
13            sp = S[p - 1]
14            p2 = p * p
15            for v in V:
16                if v < p2: break

```

```

17    S[v] -= S[v / p] - sp
18
19    return S[N]

```

6.6 Pollard Rho

Complexity: $O(\sqrt[4]{n} \log n)$

Find non-trivial factors of composite numbers with Brent's cycle detection

```

1 // Complexity:  $O(N^{1/4} \log N)$  expected time
2 // Finds a non-trivial factor of composite number  $N$  using Brent's
  cycle detection
3
4 function pollard_rho(N):
5     if N % 2 == 0: return 2
6     x = rand() % (N - 2) + 2, y = x
7     c = rand() % (N - 1) + 1, d = 1
8
9     f = (v) -> (mul_mod(v, v, N) + c) % N
10
11    while d == 1:
12        x = f(x)
13        y = f(f(y))
14        d = gcd(abs(x - y), N)
15    if d == N: return pollard_rho(N) // Retry with different
      random c
16    return d
17
18 function factorize(N, factors_list):
19     if N == 1: return
20     if is_prime(N): factors_list.append(N); return
21     d = pollard_rho(N)
22     factorize(d, factors_list)
23     factorize(N / d, factors_list)

```

6.7 Primitive Root

Complexity: $O(\text{ans} \cdot \log p)$

Find smallest primitive root g modulo prime p

```

1 // Complexity:  $O(\text{Ans} \cdot \log(P) \cdot \text{num\_factors})$  time
2 // Finds smallest primitive root modulo  $P$  (prime)
3
4 function find_primitive_root(P):
5     phi = P - 1
6     factors = unique_prime_factors(phi)
7
8     for g = 2 to P - 1:
9         is_primitive = True
10        for p in factors:
11            if power_mod(g, phi / p, P) == 1:
12                is_primitive = False
13                break
14        if is_primitive: return g
15    return -1

```

6.8 Baby-step Giant-Step

Complexity: $O(\sqrt{m} \log m)$

Find smallest x such that $a^x \equiv b \pmod{m}$ (Discrete Log)

```

1 // Complexity:  $O(\sqrt{M} \log M)$  time and  $O(\sqrt{M})$  space
2 // Finds smallest non-negative integer  $x$  such that  $a^x \equiv b \pmod{M}$ 
3
4 function baby_step_giant_step(a, b, M):

```

```

5     a %= M; b %= M
6     k = 1, add = 0, g = gcd(a, M)
7     while g > 1:
8         if b == k: return add
9         if b % g != 0: return -1 // No solution
10        b /= g; M /= g; add += 1
11        k = (k * (a / g)) % M
12        g = gcd(a, M)
13
14    m = ceil(sqrt(M))
15    table = hash_map() // Stores  $a^j \cdot m \cdot k \pmod{M} \rightarrow j$ 
16    cur = 1
17    for i = 0 to m - 1:
18        if cur not in table: table[(cur * b) % M] = i
19        cur = (cur * a) % M
20
21    base = cur //  $a^m \pmod{M}$ 
22    cur = k
23    for j = 1 to m:
24        cur = (cur * base) % M
25        if cur in table:
26            return j * m - table[cur] + add
27    return -1

```

6.9 Sqrt Mod

Complexity: $O(\log^2 p)$

Tonelli-Shanks: Solve $x^2 \equiv a \pmod{p}$

```

1 // Complexity:  $O(\log^2 P)$  expected time
2 // Tonelli-Shanks Algorithm: Solves  $x^2 \equiv a \pmod{P}$  for odd prime  $P$ 
3
4 function tonelli_shanks(a, P):
5     if a == 0: return 0
6     if power_mod(a, (P - 1) / 2, P) != 1: return -1 // No quadratic
      residue
7     if P % 4 == 3: return power_mod(a, (P + 1) / 4, P)
8
9     //  $P - 1 = Q \cdot 2^S$  with  $Q$  odd
10    S = 0, Q = P - 1
11    while Q % 2 == 0: S += 1; Q /= 2
12
13    // Find quadratic non-residue z
14    z = 2
15    while power_mod(z, (P - 1) / 2, P) == 1: z += 1
16
17    M = S, c = power_mod(z, Q, P), t = power_mod(a, Q, P), R =
      power_mod(a, (Q + 1) / 2, P)
18    while t != 1:
19        i = 1, cur = (t * t) % P
20        while cur != 1: cur = (cur * cur) % P; i += 1
21        b = power_mod(c, 1 << (M - i - 1), P)
22        M = i; c = (b * b) % P; t = (t * c) % P; R = (R * b) % P
23    return R

```

6.10 Extended GCD

Complexity: $O(\log \min(a, b))$

Find x, y that satisfy the equation $ax + by = \gcd(a, b)$

```

1 // Complexity:  $O(\log(\min(a, b)))$ 
2 // Finds  $x, y$  such that  $a*x + b*y = \gcd(a, b)$ 
3
4 function extgcd(a, b):
5     if b == 0: return (g = a, x = 1, y = 0)
6     res = extgcd(b, a % b)
7     return (g = res.g, x = res.y, y = res.x - (a / b) * res.y)

```

```

8
9 function mod_inverse(a, m):
10     res = extgcd(a, m)
11     if res.g != 1: return -1 // Inverse does not exist
12     return (res.x % m + m) % m

```

6.11 Diophantine

Complexity: $O(\log \min(a, b))$

General solutions to linear Diophantine equation $ax + by = c$

```

1 // Complexity: O(log(min(a, b)))
2 // Solves a*x + b*y = c. General solutions: x = x0 + k*(b/g), y =
  y0 - k*(a/g)
3
4 function solve_diophantine(a, b, c):
5     res = extgcd(abs(a), abs(b))
6     g = res.g
7     if c % g != 0: return (has_sol = False, 0, 0)
8     x0 = res.x * (c / g)
9     y0 = res.y * (c / g)
10    if a < 0: x0 = -x0
11    if b < 0: y0 = -y0
12    return (has_sol = True, x0, y0, g)

```

6.12 Stern-Brocot Tree

Complexity: $O(\log(p + q))$

Fractions in reduced forms, tree encoding/decoding, LCA

```

1 // Complexity: O(log(p + q))
2 // Binary search on Stern-Brocot tree to encode/decode positive
  reduced fractions p / q
3
4 function encode_path(p, q):
5     path = []
6     L_num = 0, L_den = 1; R_num = 1, R_den = 0
7     while True:
8         M_num = L_num + R_num, M_den = L_den + R_den
9         if p * M_den == q * M_num: break
10        if p * M_den < q * M_num: // Go Left
11            k = (q * L_num - p * L_den) / (p * R_den - q * R_num)
12        // accelerate steps
13        k = max(1, k)
14        path.append({'L', k})
15        R_num += k * L_num; R_den += k * L_den
16    else: // Go Right
17        k = (p * R_den - q * R_num) / (q * L_num - p * L_den)
18        k = max(1, k)
19        path.append({'R', k})
20        L_num += k * R_num; L_den += k * R_den
21    return path

```

6.13 Multiplicative Functions & Möbius Inversion

Dirichlet Convolution, Möbius Inversion, GCD/LCM summations

• **Euler Phi & Möbius:** $\phi(n) = n \prod_{p|n} (1 - 1/p)$, $\sum_{d|n} \phi(d) = n$, $\sum_{d|n} \mu(d) = [n = 1]$.

• **Dirichlet & Inversion:** $(f * g)(n) = \sum_{d|n} f(d)g(n/d)$.

$$f(n) = \sum_{d|n} g(d) \iff g(n) = \sum_{d|n} \mu(d)f(n/d)$$

• **Count coprime pairs:**

$$\sum_{i=1}^n \sum_{j=1}^m [\gcd(i, j) = 1] = \sum_{d=1}^{\min(n, m)} \mu(d) \left\lfloor \frac{n}{d} \right\rfloor \left\lfloor \frac{m}{d} \right\rfloor$$

• **Sum of GCD over all pairs:**

$$\sum_{i=1}^n \sum_{j=1}^m \gcd(i, j) = \sum_{g=1}^{\min(n, m)} \phi(g) \left\lfloor \frac{n}{g} \right\rfloor \left\lfloor \frac{m}{g} \right\rfloor$$

• **Sum of LCM over all pairs** ($S(x) = \frac{x(x+1)}{2}$):

$$\sum_{i \leq n, j \leq m} \text{lcm}(i, j) = \sum_g g \sum_d d^2 \mu(d) S\left(\left\lfloor \frac{n}{gd} \right\rfloor\right) S\left(\left\lfloor \frac{m}{gd} \right\rfloor\right)$$

6.14 Combinatorics Formulas

Binomial identities, Hockey-stick, Vandermonde, Inversion, Stars and bars, Catalan, Stirling, Cayley, LGV

• **Binomial Identities:** $\binom{n}{k} = \binom{n}{n-k} = \binom{n-1}{k-1} + \binom{n-1}{k}$; $k \binom{n}{k} = n \binom{n-1}{k-1}$; $\frac{1}{k+1} \binom{n}{k} = \frac{1}{n+1} \binom{n+1}{k+1}$; $\binom{n}{k} \binom{k}{m} = \binom{n}{m} \binom{n-m}{k-m}$; Upper negation: $\binom{-n}{k} = (-1)^k \binom{n+k-1}{k}$.

• **Binomial Summations:** $\sum_{k=0}^n \binom{n}{k} = 2^n$; $\sum_{k=0}^n (-1)^k \binom{n}{k} = [n = 0]$; $\sum_k \text{even} \binom{n}{k} = \sum_k \text{odd} \binom{n}{k} = 2^{n-1}$; $\sum_{k=0}^n k \binom{n}{k} = n 2^{n-1}$; $\sum_{k=0}^n k^2 \binom{n}{k} = n(n+1) 2^{n-2}$; $\sum_{k=0}^n \frac{1}{k+1} \binom{n}{k} = \frac{2^{n+1}-1}{n+1}$.

• **Hockey-Stick & Vandermonde:** $\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$; $\sum_{k=0}^n \binom{n+k}{k} = \binom{n+1}{k+1}$; $\sum_{k=0}^n \binom{n}{k} \binom{n-k}{r-k} = \binom{n+1}{r}$ $\implies \sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$.

• **Binomial Inversion & Differences:** $f(n) = \sum_{k=0}^n \binom{n}{k} g(k) \iff g(n) = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f(k)$; Upper: $f(n) = \sum_{k=n}^N \binom{k}{n} g(k) \iff g(n) = \sum_{k=n}^N (-1)^{k-n} \binom{k}{n} f(k)$; $\sum_{k=0}^n (-1)^{n-k} \binom{n}{k} k^m = n!$ $\left\{ \begin{smallmatrix} m \\ n \end{smallmatrix} \right\}$ (0 if $m < n$, $n!$ if $m = n$).

• **Stars and Bars:** $x_i \geq 0 \implies \binom{n+k-1}{k-1}$; $x_i \geq 1 \implies \binom{n-1}{k-1}$; $x_i \geq c_i \implies n' = n - \sum c_i$.

• **Derangements:** $D_0 = 1, D_1 = 0, D_n = (n-1)(D_{n-1} + D_{n-2}) = n D_{n-1} + (-1)^n \approx [n!/e]$.

• **Catalan Numbers** $C_n = \frac{1}{n+1} \binom{2n}{n}$: Valid bracketings, Full binary trees with $n+1$ leaves, Triangulations of convex $(n+2)$ -gon, Dyck paths not crossing $y = x$, 1-Stack permutations.

• **Reflection Principle:** Paths $(0, 0) \rightarrow (n, m)$ not touching $y = x + k$ ($k > 0$): $\binom{n+m}{m} - \binom{n+m}{m-k}$.

• **Balloting Theorem:** A has a votes, B has b ($a > b$). Prob A strictly leads throughout: $\frac{a-b}{a+b}$.

• **Stirling Numbers of 1st & 2nd Kind:** $\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] = \left[\begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right] + (n-1) \left[\begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right]$, $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right\} + k \left\{ \begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right\}$. $x^n = \sum_{k=0}^n \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} x^{\underline{k}} \implies \sum_{i=1}^n i^k = \sum_{j=0}^k \left\{ \begin{smallmatrix} k \\ j \end{smallmatrix} \right\} j! \binom{n+1}{j+1}$.

• **PIE:** $|\bigcup_{i=1}^n A_i| = \sum_{\emptyset \neq J} (-1)^{|J|-1} |\bigcap_{j \in J} A_j|$; Exactly r sets: $\sum_{m=r}^n (-1)^{m-r} \binom{m}{r} \sum_{|X|=m} |\bigcap_{i \in X} A_i|$.

• **Lucas' Theorem:** $\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod{p}$ (base- p digits).

• **Tree Counting & Cayley:** n^{n-2} labeled trees, Forest of k rooted trees kn^{n-k-1} , Matrix Tree Theorem ($\det(L_i)$), LGV Lemma ($\det(M)$).

6.15 Generating Functions & Taylor Series

OGF, EGF, Taylor expansions, Simpson numerical integration

• OGF: $A(x) = \sum a_n x^n$; EGF: $A(x) = \sum a_n \frac{x^n}{n!}$.

• $\frac{1}{1-x} = \sum_{n \geq 0} x^n$, $\frac{1}{(1-x)^k} = \sum_{n \geq 0} \binom{n+k-1}{k-1} x^n$.

• $e^x = \sum_{n \geq 0} \frac{x^n}{n!}$, $\ln(1-x) = -\sum_{n \geq 1} \frac{x^n}{n}$, $(1+x)^\alpha = \sum_{n \geq 0} \binom{\alpha}{n} x^n$.

• **Simpson's Rule:** $\int_a^{a+2h} f(x) dx \approx \frac{h}{3} (f(a) + 4f(a+h) + f(a+2h))$.

6.16 Linear Recurrences (Berlekamp-Massey)

Complexity: $O(N^2)$ find recurrence, $O(N^2 \log K)$ compute k -th term

Find minimal linear recurrence from first $2N$ terms and compute K -th term via $x^K \bmod P(x)$

```

1 // Berlekamp-Massey: Finds shortest linear recurrence C: s[i] = \
  sum_{j=0}^{L-1} C[j] * s[i-1-j]
2 // linearRec: Computes K-th term (0-indexed) in O(N^2 log K) via x^K
  mod P(x)
3 vector<int> berlekampMassey(const vector<int>& s) {
4     int n = s.size(), L = 0, m = 0;
5     vector<int> C(n, 0), B(n, 0), T;
6     C[0] = B[0] = 1;
7     long long b = 1;
8     for (int i = 0; i < n; ++i) {
9         ++m;
10        long long d = s[i] % MOD;
11        for (int j = 1; j <= L; ++j) d = (d + 1LL * C[j] * s[i - j]
12        ) % MOD;
13        if (1d) continue;
14        T = C;
15        long long coef = d * power(b, MOD - 2) % MOD;
16        for (int j = m; j < n; ++j) C[j] = (C[j] - coef * B[j - m]
17        % MOD + MOD) % MOD;
18        if (2 * L > i) continue;
19        L = i + 1 - L; B = T; b = d; m = 0;
20    }
21    C.resize(L + 1); C.erase(C.begin());
22    for (int& x : C) x = (MOD - x) % MOD;
23    return C; // recurrence coefficients
24 }
25
26 int linearRec(const vector<int>& S, const vector<int>& tr, long
  long k) {
27     int n = tr.size();
28     if (n == 0) return 0;
29     if (k < (int)S.size()) return (S[k] % MOD + MOD) % MOD;
30     auto combine = [&](const vector<int>& a, const vector<int>& b)
31     {
32         vector<int> res(2 * n + 1, 0);
33         for (int i = 0; i <= n; ++i) for (int j = 0; j <= n; ++j)
34             res[i + j] = (res[i + j] + 1LL * a[i] * b[j]) % MOD;
35         for (int i = 2 * n; i > n; --i) for (int j = 0; j < n; ++j)
36             res[i - 1 - j] = (res[i - 1 - j] + 1LL * res[i] * tr[j]
37             ) % MOD;
38         res.resize(n + 1);

```

```

35     return res;
36 };
37 vector<int> pol(n + 1, 0), e(n + 1, 0);
38 pol[0] = e[1] = 1;
39 for (++k; k; k /= 2) {
40     if (k % 2) pol = combine(pol, e);
41     e = combine(e, e);
42 }
43 long long res = 0;
44 for (int i = 0; i < n; i++) res = (res + 1LL * pol[i + 1] * S[i]
45     ]) % MOD;
46 return (res + MOD) % MOD;
47 }

```

6.17 FWHT

Complexity: $\mathcal{O}(n \log n)$

Fast Walsh-Hadamard Transform (XOR, AND, OR Convolution)

```

1 // Complexity:  $\mathcal{O}(N \log N)$  where  $N = 2^k$ 
2 // Fast Walsh-Hadamard Transform for bitwise XOR convolution ( $C[k]$ 
3 // =  $\sum_{i \oplus j = k} A[i] * B[j]$ )
4 function fwht_xor(a, invert):
5     n = len(a)
6     len_step = 1
7     while len_step < n:
8         for i = 0 to n - 1 step 2 * len_step:
9             for j = 0 to len_step - 1:
10                u = a[i + j], v = a[i + len_step + j]
11                a[i + j] = (u + v) % MOD
12                a[i + len_step + j] = (u - v + MOD) % MOD
13            len_step *= 2
14     if invert:
15         inv_n = power_mod(n, MOD - 2, MOD)
16         for i = 0 to n - 1: a[i] = (a[i] * inv_n) % MOD

```

6.18 Lagrange

Complexity: $\mathcal{O}(n)$ evaluation

Lagrange polynomial interpolation at point X on $n + 1$ equidistant points $y_0, \dots, y_n$

```

1 // Complexity:  $\mathcal{O}(N)$  evaluation at point  $X$  for consecutive points
2 // Formula:  $f(X) = \sum_{i=0}^{n-1} y_i \prod_{j \neq i} \frac{X - j}{i - j}$ 
3
4 function lagrange_eval_consecutive(y[0..n], X):
5     if X <= n: return y[X]
6     pref[0..n], suff[0..n]
7     pref[0] = X, suff[n] = (X - n) % MOD
8     for i = 1 to n: pref[i] = (pref[i-1] * (X - i)) % MOD
9     for i = n - 1 down to 0: suff[i] = (suff[i+1] * (X - i)) % MOD
10
11     ans = 0
12     for i = 0 to n:
13         num = (1 if i == 0 else pref[i-1]) * (1 if i == n else suff[i+1]) % MOD
14         den = (fact[i] * fact[n - i]) % MOD
15         term = (y[i] * num % MOD) * inv(den) % MOD
16         if (n - i) % 2 == 1: ans = (ans - term + MOD) % MOD
17         else: ans = (ans + term) % MOD
18     return ans

```

6.19 NTT (Number Theoretic Transform)

Complexity: $\mathcal{O}(n \log n)$

Iterative NTT with bit-reversal under $\text{MOD} = 998244353$ ($G = 3$)

```

1 const int MOD = 998244353, G = 3;
2 long long power(long long a, long long b) {
3     long long res = 1;
4     for (a %= MOD; b; b >>= 1, a = a * a % MOD) if (b & 1) res =
5         res * a % MOD;
6     return res;
7 long long modInverse(long long n) { return power(n, MOD - 2); }
8
9 void ntt(vector<int>& a, bool invert) {
10     int n = a.size();
11     for (int i = 1, j = 0; i < n; i++) {
12         int bit = n >> 1;
13         for (; j & bit; bit >>= 1) j ^= bit;
14         j ^= bit;
15         if (i < j) swap(a[i], a[j]);
16     }
17     for (int len = 2; len <= n; len <= 1) {
18         long long wlen = power(G, (MOD - 1) / len);
19         if (invert) wlen = modInverse(wlen);
20         for (int i = 0; i < n; i += len) {
21             long long w = 1;
22             for (int j = 0; j < len / 2; j++) {
23                 long long u = a[i + j], v = a[i + j + len / 2] * w
24                     % MOD;
25                 a[i + j] = (u + v >= MOD ? u + v - MOD : u + v);
26                 a[i + j + len / 2] = (u - v < 0 ? u - v + MOD : u -
27                     v);
28                 w = w * wlen % MOD;
29             }
30         }
31     if (invert) {
32         long long inv_n = modInverse(n);
33         for (int& x : a) x = x * inv_n % MOD;
34     }
35 }
36 vector<int> multiply(vector<int> a, vector<int> b) {
37     int sz = 1, need = a.size() + b.size() - 1;
38     while (sz < need) sz <= 1;
39     a.resize(sz); b.resize(sz);
40     ntt(a, false); ntt(b, false);
41     for (int i = 0; i < sz; i++) a[i] = 1LL * a[i] * b[i] % MOD;
42     ntt(a, true);
43     a.resize(need);
44     return a;
45 }

```

6.20 FFT (Fast Fourier Transform)

Complexity: $\mathcal{O}(n \log n)$

Cooley-Tukey FFT with complex numbers for real/arbitrary convolution

```

1 using cd = complex<double>;
2 const double PI = acos(-1);
3
4 void fft(vector<cd>& a, bool invert) {
5     int n = a.size();
6     for (int i = 1, j = 0; i < n; i++) {
7         int bit = n >> 1;
8         for (; j & bit; bit >>= 1) j ^= bit;

```

```

9         j ^= bit;
10        if (i < j) swap(a[i], a[j]);
11    }
12    for (int len = 2; len <= n; len <= 1) {
13        double ang = 2 * PI / len * (invert ? -1 : 1);
14        cd wlen(cos(ang), sin(ang));
15        for (int i = 0; i < n; i += len) {
16            cd w(1);
17            for (int j = 0; j < len / 2; j++) {
18                cd u = a[i + j], v = a[i + j + len / 2] * w;
19                a[i + j] = u + v;
20                a[i + j + len / 2] = u - v;
21                w *= wlen;
22            }
23        }
24    }
25    if (invert) for (cd& x : a) x /= n;
26 }
27
28 vector<long long> multiply_fft(const vector<int>& a, const vector<
29     int>& b) {
30     vector<cd> fa(a.begin(), a.end()), fb(b.begin(), b.end());
31     int sz = 1, need = a.size() + b.size() - 1;
32     while (sz < need) sz <= 1;
33     fa.resize(sz); fb.resize(sz);
34     fft(fa, false); fft(fb, false);
35     for (int i = 0; i < sz; i++) fa[i] *= fb[i];
36     fft(fa, true);
37     vector<long long> res(need);
38     for (int i = 0; i < need; i++) res[i] = round(fa[i].real());
39     return res;
40 }

```

6.21 Formal Power Series (FPS) Formulas

Key formulas for polynomial operations via Newton's method & convolution

• **Inverse** ($Q = P^{-1}$): $Q_k \equiv Q_{k-1}(2 - PQ_{k-1}) \pmod{x^{2^k}}$, base $Q_0 = P(0)^{-1}$.

• **Derivative & Integral**: $P'(x) = \sum ia_i x^{i-1}$, $\int P(x) = \sum \frac{a_i}{i+1} x^{i+1}$.

• **Logarithm**: $\ln P(x) = \int \frac{P'(x)}{P(x)} dx \pmod{x^n}$ (requires $P(0) = 1$).

• **Exponential**: $Q_k \equiv Q_{k-1}(1 - \ln Q_{k-1} + P) \pmod{x^{2^k}}$ (requires $P(0) = 0$).

• **Square Root**: $Q_k \equiv \frac{1}{2}(Q_{k-1} + PQ_{k-1}^{-1}) \pmod{x^{2^k}}$ ($P(0)$ is quadratic residue).

• **Power**: $P(x)^k = \exp(k \ln P(x))$. If $P(x) = cx^d(1 + Q(x))$, $P(x)^k = c^k x^{kd} \exp(k \ln(1 + Q(x)))$.

• **Division with Remainder** ($A = BQ + R$): With $\text{rev}_k(P) = x^k P(1/x)$:

$\text{rev}_{n-m}(Q) \equiv \text{rev}_n(A) \cdot (\text{rev}_m(B))^{-1} \pmod{x^{n-m+1}}$, $R = A - BQ$

• **Taylor Shift** $P(x+c)$: $P(x+c) = \sum_{k=0}^n \frac{x^k}{k!} \sum_{i=k}^n (i! a_i) \frac{c^{i-k}}{(i-k)!}$. Convolute reversed $A_i = i! a_i$ with $B_j = \frac{c^j}{j!}$.

• **Any-Mod Convolution (MTT)**: Split $A = A_1 B + A_0$, $B = \lceil \sqrt{M} \rceil \approx 32000$. Compute $(A_1 B_1), (A_1 B_0 + A_0 B_1), (A_0 B_0)$ with 4 FFTs.

6.22 Linear Algebra (Gauss)

Complexity: $\mathcal{O}(n^3)$

Gaussian Elimination: Matrix Determinant, Matrix Rank, System of Linear Equations

```
1 // Complexity:  $\mathcal{O}(N^3)$  or  $\mathcal{O}(N^3 / 64)$  with Bitset
2 // Gaussian Elimination: Determinant, Matrix Inversion, System of
  Linear Equations  $Ax = b$ 
3
4 function gauss(A[N][M], b[N]):
5     // Augmented matrix [A | b]
6     rank = 0, det = 1
7     for col = 0 to M - 1:
8         pivot = find_row_with_max_abs(col, from_row = rank)
9         if abs(A[pivot][col]) < EPS:
10             det = 0; continue
11         swap(row[rank], row[pivot])
12         if rank != pivot: det = -det
13         det *= A[rank][col]
14
15         // Eliminate column entries below and above pivot
16         for r = 0 to N - 1:
17             if r != rank and abs(A[r][col]) > EPS:
18                 factor = A[r][col] / A[rank][col]
19                 row[r] -= factor * row[rank]
20         rank += 1
21 return (rank, det, solution = [b[i] / A[i][i] for i in 0..rank - 1])
```

6.23 XOR Linear Basis

Complexity: $\mathcal{O}(B)$ insert/query ($B \leq 60$)

XOR Linear Basis: Find Max/Min/k-th XOR subset, span of 2^{size} distinct values

```
1 // Complexity:  $\mathcal{O}(B)$  per insert,  $B = 60$  bits
2 // Recognition: Subset XOR max/min/k-th, distinct subset XOR sums,
  cycle XOR in graphs
3
4 struct XORBasis:
5     basis[B] = {0}, sz = 0
6
7     function insert(mask):
8         for i = B - 1 down to 0:
9             if not (mask & (1 << i)): continue
10            if not basis[i]:
11                basis[i] = mask; sz += 1
12            return True
13            mask ^= basis[i]
14        return False // mask is linearly dependent
15
16     function query_max():
17         res = 0
18         for i = B - 1 down to 0:
19             res = max(res, res ^ basis[i])
20        return res
21
22     function query_kth(k):
23         // Rebuild reduced row echelon form first:
24         // for each basis[i], eliminate 1-bits in basis[j] ( $j > i$ )
25         // If 0 is in span (insert failed at least once), decrement
26         k if  $k \geq 1$ 
27         res = 0, total_distinct = 1 << sz
28         if k > total_distinct: return -1
29         for i = 0 to sz - 1:
30             if k & (1 << i): res ^= reduced_basis[i]
31         return res
```

6.24 Burnside Lemma & Pólya Theorem

Count distinct equivalence classes under group actions, Necklaces, Bracelets

• $|Classes| = \frac{1}{|G|} \sum_{g \in G} |Fix(g)| = \frac{1}{|G|} \sum_{g \in G} k^{c(g)}$.

• **Necklace (n rotations only):** $N(n, k) = \frac{1}{n} \sum_{d|n} \phi(d) k^{n/d}$.

• **Bracelet (Rotations + Reflections):**

– n odd: $B(n, k) = \frac{1}{2} N(n, k) + \frac{1}{2} k^{(n+1)/2}$.

– n even: $B(n, k) = \frac{1}{2} N(n, k) + \frac{1}{4} (k^{n/2} + k^{n/2+1})$.

6.25 Game Strategy & State Reasoning

Complexity: $\mathcal{O}(V + E)$

Winning/Losing state axioms, Retrograde analysis, Symmetry, Parity, Sprague-Grundy MEX, Nim variants

```
1 const int DRAW = 0, WIN = 1, LOSE = -1;
2
3 vector<int> retrograde_analysis(int n, const vector<vector<int>>&
  adj) {
4     vector<int> out_deg(n), res(n, DRAW);
5     vector<vector<int>> radj(n);
6     queue<int> q;
7
8     for (int u = 0; u < n; ++u) {
9         out_deg[u] = adj[u].size();
10        for (int v : adj[u]) radj[v].push_back(u);
11        if (out_deg[u] == 0) {
12            res[u] = LOSE;
13            q.push(u);
14        }
15    }
16
17    while (!q.empty()) {
18        int u = q.front(); q.pop();
19        for (int p : radj[u]) {
20            if (res[p] != DRAW) continue;
21            if (res[u] == LOSE) {
22                res[p] = WIN; q.push(p);
23            } else if (res[u] == WIN) {
24                if (--out_deg[p] == 0) {
25                    res[p] = LOSE; q.push(p);
26                }
27            }
28        }
29    }
30    return res;
31 }
32
33 int calc_mex(const vector<int>& vals) {
34     int m = vals.size();
35     vector<bool> seen(m + 1, false);
36     for (int x : vals) if (x <= m) seen[x] = true;
37     for (int i = 0; i <= m; ++i) if (!seen[i]) return i;
38     return m + 1;
39 }
40
41 int get_grundy(int u, vector<int>& g, const vector<vector<int>>&
  adj) {
42     if (g[u] != -1) return g[u];
43     vector<int> nxt;
44     for (int v : adj[u]) nxt.push_back(get_grundy(v, g, adj));
45     return g[u] = calc_mex(nxt);
46 }
```

• **Winning ($\mathcal{W}$ / N-pos) vs Losing ($\mathcal{L}$ / P-pos) State Axioms:**

– **Terminal State:** Normal play (no moves $\implies \mathcal{L}$); Misère play (no moves $\implies \mathcal{W}$).

– **Winning State $\mathcal{W}$:** $\exists v \in \text{next}(u)$ such that $v \in \mathcal{L}$ (at least ONE move leads to an $\mathcal{L}$ -state).

– **Losing State $\mathcal{L}$:** $\forall v \in \text{next}(u)$, $v \in \mathcal{W}$ (ALL moves lead to $\mathcal{W}$ -states; or no moves exist).

– **Optimal Play:** From $\mathcal{W}$, always choose a move to $\mathcal{L}$. Opponent from $\mathcal{L}$ is forced to move back to $\mathcal{W}$.

• Problem-Solving Deduction Workflow:

– **Retrograde BFS ($\mathcal{O}(V + E)$):** Propagate outcomes backward from terminal states using out-degree counters; unvisited states in cycles are **DRAW**.

– **Symmetry / Mirroring:** If P2 can mirror P1's moves across center/axes, P2 wins. If P1 can make a central move leaving symmetric halves, P1 wins.

– **Invariants & Parity:** Track $I(S) \bmod 2$ (sum, components, inversions). Control parity to force terminal state on opponent.

– **Pairing Strategy:** Partition board into disjoint matched pairs (a_i, b_i) . When P1 takes a_i , P2 immediately takes b_i .

– **Small-case Pattern Finding:** Simulate for $N \leq 20$, compute SG values $g(n)$, find periodicity $g(n) = g(n \bmod T)$ or bit conditions.

• Sprague-Grundy Theorem (Independent Subgames):

– $g(G) = \bigoplus g(G_i)$, where $g(u) = \text{mex}(\{g(v) \mid u \rightarrow v\})$. Win $\iff g(G) \neq 0$.

– **Tree Hackenbush (Colon Principle):** Branch node $g(u) = \bigoplus_{v \in \text{children}} (g(v) + 1)$.

• Classic Nim Variations:

– **Standard Nim:** XOR sum $S = \bigoplus a_i$. Win $\iff S \neq 0$.

– **Bounded Subtraction Nim** (take 1..k): $g(a) = a \bmod (k + 1) \implies \text{Win} \iff \bigoplus (a_i \bmod (k + 1)) \neq 0$.

– **Moore's Nim_k** (take from $\leq k$ piles): Lose $\iff \forall \text{ bit position } b, \sum_i \text{bit}_b(a_i) \equiv 0 \pmod{k + 1}$.

– **Staircase Nim:** Equivalent to standard Nim on odd-indexed stairs $(1, 3, 5, \dots)$.

– **Misère Nim** (last move LOSES):

* All piles ≤ 1 : Win $\iff$ count of size-1 piles is EVEN ($S = 0$).

* At least one pile > 1 : Identical to standard Nim (Win $\iff S \neq 0$).

6.26 Probability & Expected Value

Linearity of expectation, Indicator variables, Coupon collector, Inequalities

• **Linearity of Expectation:** $E[X_1 + \dots + X_n] = E[X_1] + \dots + E[X_n]$ ($\forall X_i$).

• **Indicator Variables:** $E[X] = \sum P(A_i)$.

• **Geometric Distribution:** $E[X] = 1/p$, $\text{Var}(X) = (1 - p)/p^2$.

• **Coupon Collector:** $E[T] = n \sum_{i=1}^n \frac{1}{i} = nH_n \approx n(\ln n + \gamma)$ ($\gamma \approx 0.5772$).

• **Markov:** $P(X \geq a) \leq \frac{E[X]}{a}$; **Chebyshev:** $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$.

6.27 Bignum

Complexity: $O(n^2)$
Extreme Bignum Template

```
1 using bignum = vector<int>;
2
3 bignum operator + (const bignum &a, const bignum &b){
4     bignum res;
5     int carry = 0;
6     for(int i = 0; i < max(b.size(), a.size()); i++){
7         int c1 = (i < sz(a)) ? a[i] : 0;
8         int c2 = (i < sz(b)) ? b[i] : 0;
9         int c = (c1 + c2 + carry);
10        res.pb(c % 10);
11        carry = c / 10;
12    }
13    while(carry) {
14        res.pb(carry % 10);
15        carry /= 10;
16    }
17    return res;
18 }
19 bignum operator * (const bignum &a, int k){
20     bignum res;
21     int carry = 0;
22     for(int i = 0; i < a.size(); i++){
23         int c = (a[i] * k + carry);
24         res.pb(c % 10);
25         carry = c / 10;
26     }
27     while(carry) {
28         res.pb(carry % 10);
29         carry /= 10;
30     }
31     return res;
32 }
```

7 Geometry

7.1 Primitive Geometry

Complexity: $O(1)$ per operation
Core 2D primitives: Points, Vectors, Orientation, Line Intersections, Projections

```
1 // Complexity: O(1) per primitive operation
2 // Core 2D Geometry: Points, Vectors, Lines, Circles
3
4 struct Point:
5     x, y
6     operator + (p) => Point(x + p.x, y + p.y)
7     operator - (p) => Point(x - p.x, y - p.y)
8     operator * (k) => Point(x * k, y * k)
9     dot(p) => x * p.x + y * p.y
10    cross(p) => x * p.y - y * p.x // > 0 if p is counter-clockwise
11    (left turn)
12    dist2() => x * x + y * y
13    dist() => sqrt(dist2())
14
15 function orientation(a, b, c):
16     val = (b - a).cross(c - a)
17     if abs(val) < EPS: return 0 // Collinear
18     return 1 if val > 0 else -1 // 1: CCW (Left), -1: CW (Right)
19
20 function line_intersection(p1, p2, q1, q2):
21     // Line 1 through p1, p2; Line 2 through q1, q2
22     cross_val = (p2 - p1).cross(q2 - q1)
```

```
22     if abs(cross_val) < EPS: return NULL // Parallel or coincident
23     t = (q1 - p1).cross(q2 - q1) / cross_val
24     return p1 + (p2 - p1) * t
```

7.2 Convex Hull

Complexity: $O(n \log n)$
Monotone Chain / Graham Scan: Smallest convex polygon enclosing point set

```
1 // Complexity: O(N log N) time, O(N) space
2 // Monotone Chain algorithm: Constructs convex hull in counter-
3 // clockwise order
4
5 function convex_hull(points P):
6     sort(P) // Lexicographically by x, then y
7     lower = [], upper = []
8
9     for p in P:
10        while len(lower) >= 2 and orientation(lower[len-2], lower[
11        len-1], p) <= 0:
12            lower.pop()
13        lower.append(p)
14
15    for p in reversed(P):
16        while len(upper) >= 2 and orientation(upper[len-2], upper[
17        len-1], p) <= 0:
18            upper.pop()
19        upper.append(p)
20
21    lower.pop(); upper.pop()
22    return lower + upper
```

7.3 Rotating Calipers

Complexity: $O(n)$
Rotating Calipers: Polygon diameter (farthest pair), Minimum bounding rectangle

```
1 // Complexity: O(N) on convex hull in CCW order
2 // Recognition: Maximum pairwise distance (diameter), Minimum
3 // bounding box, Max triangle area
4
5 function convex_diameter(poly CH):
6     n = len(CH)
7     if n <= 1: return 0
8     if n == 2: return dist(CH[0], CH[1])
9
10    max_d2 = 0, k = 1
11    // Find initial farthest point k from edge (CH[0], CH[1])
12    while triangle_area(CH[n-1], CH[0], CH[(k + 1) % n]) >
13        triangle_area(CH[n-1], CH[0], CH[k]):
14        k = (k + 1) % n
15
16    for i = 0 to n - 1:
17        // Advance caliper pointer k while distance to edge (CH[i],
18        // CH[i+1]) increases
19        next_i = (i + 1) % n
20        while triangle_area(CH[i], CH[next_i], CH[(k + 1) % n]) >=
21            triangle_area(CH[i], CH[next_i], CH[k]):
22            k = (k + 1) % n
23        max_d2 = max(max_d2, dist2(CH[i], CH[k]))
24        max_d2 = max(max_d2, dist2(CH[next_i], CH[k]))
25    return sqrt(max_d2)
```

7.4 Minkowski Sum

Complexity: $O(|P| + |Q|)$
Minkowski Sum of two convex polygons. Min polygon distance:
 $\text{dist}(P, Q) = \text{dist}(O, P - Q)$

```
1 // Complexity: O(|P| + |Q|) time
2 // Computes Minkowski Sum P + Q = {p + q | p in P, q in Q} for
3 // convex polygons in CCW order
4
5 function minkowski_sum(poly P, poly Q):
6     // Reorder vertices starting at bottom-left
7     rotate_to_min_vertex(P); rotate_to_min_vertex(Q)
8     P.append(P[0]); P.append(P[1])
9     Q.append(Q[0]); Q.append(Q[1])
10
11    i = 0, j = 0, result = []
12    while i < len(P) - 2 or j < len(Q) - 2:
13        result.append(P[i] + Q[j])
14        cross_val = (P[i+1] - P[i]).cross(Q[j+1] - Q[j])
15        if cross_val >= 0 and i < len(P) - 2: i += 1
16        if cross_val <= 0 and j < len(Q) - 2: j += 1
17    return result
```

7.5 Point in Convex Polygon

Complexity: $O(\log n)$
Test if point lies inside convex polygon via polar angle binary search from P_0

```
1 // Complexity: O(log N) time per query on convex polygon in CCW
2 // order
3 // Recognition: Fast point inclusion testing in convex polygon
4
5 function point_in_convex_polygon(poly P, Point pt):
6     n = len(P)
7     // Check if pt is within the sector between P[0]->P[1] and P
8     // [0]->P[n-1]
9     if orientation(P[0], P[1], pt) < 0 or orientation(P[0], P[n -
10     1], pt) > 0:
11         return OUTSIDE
12
13    // Binary search to find triangle P[0] - P[mid] - P[mid+1]
14    // containing pt
15    low = 1, high = n - 2
16    while low <= high:
17        mid = (low + high) / 2
18        if orientation(P[0], P[mid], pt) >= 0:
19            pos = mid; low = mid + 1
20        else:
21            high = mid - 1
22
23    // Check if pt is inside triangle P[0] - P[pos] - P[pos+1]
24    val = orientation(P[pos], P[pos + 1], pt)
25    if val > 0: return INSIDE
26    if val == 0: return ON_BOUNDARY
27    return OUTSIDE
```

7.6 Line Hull Intersection

Complexity: $O(\log n)$
Binary search extreme vertices along normal vector to find intersection points

```
1 // Complexity: O(log N) time per line query
2 // Finds intersection of line L with convex hull CH using binary
3 // search
```

```

4 function extreme_vertex(poly CH, direction_vector V):
5     // Binary search on ternary search of dot product (P[i] . V)
6     low = 0, high = len(CH) - 1
7     while high - low > 2:
8         m1 = low + (high - low) / 3
9         m2 = high - (high - low) / 3
10        if CH[m1].dot(V) < CH[m2].dot(V): low = m1
11        else: high = m2
12    return argmax_{i in [low..high]} (CH[i].dot(V))
13
14 function line_hull_intersect(poly CH, Line L):
15     n1 = extreme_vertex(CH, L.normal)
16     n2 = extreme_vertex(CH, -L.normal)
17     if L.side(CH[n1]) * L.side(CH[n2]) > 0:
18         return (intersects = False, [])
19     // Binary search on both chains between n1 and n2 to find exact
20     // crossing edges
21     return (intersects = True, [binary_search_cross(CH, n1, n2, L),
22         binary_search_cross(CH, n2, n1, L)])

```

7.7 Polygon Raycast

Complexity: $O(n)$

Horizontal raycasting point-in-polygon test for arbitrary simple polygons

```

1 // Complexity: O(N) time for point in polygon, O(log N) for convex
2 // Ray-Casting algorithm: Cast horizontal ray from P to +infinity,
3 // count edge crossings
4
5 function point_in_polygon(poly P, Point pt):
6     crossings = 0, n = len(P)
7     for i = 0 to n - 1:
8         a = P[i], b = P[(i + 1) % n]
9         if on_segment(a, b, pt): return ON_BOUNDARY
10        if (a.y > pt.y) != (b.y > pt.y):
11            // Calculate x-coordinate of intersection with
12            // horizontal line y = pt.y
13            x_cross = a.x + (pt.y - a.y) * (b.x - a.x) / (b.y - a.y)
14            if pt.x < x_cross:
15                crossings += 1
16    return INSIDE if (crossings % 2 == 1) else OUTSIDE

```

7.8 Half-plane Intersection

Complexity: $O(n \log n)$

Intersection of half-planes forming convex polygon via Deque

```

1 // Complexity: O(N log N) time, O(N) space
2 // Computes convex polygon formed by intersection of halfplanes (
3 // directed lines with valid side on left)
4
5 struct Halfplane:
6     p, pq // point p and direction vector pq
7
8 function halfplane_intersection(halfplanes H):
9     sort(H, by polar_angle)
10    deque<Halfplane> dq; deque<Point> pts
11
12    for h in H:
13        while len(pts) > 0 and not h.contains(pts.back()):
14            pts.pop_back(); dq.pop_back()
15        while len(pts) > 0 and not h.contains(pts.front()):
16            pts.pop_front(); dq.pop_front()
17        if len(dq) > 0:

```

```

17        pts.push_back(line_intersection(dq.back(), h))
18        dq.push_back(h)
19
20    while len(pts) > 0 and not dq.front().contains(pts.back()):
21        pts.pop_back(); dq.pop_back()
22    if len(dq) < 3: return [] // Empty intersection
23    pts.push_back(line_intersection(dq.back(), dq.front()))
24    return pts // Vertices of convex polygon

```

7.9 Smallest Enclosing Circle

Complexity: $O(n)$

Welzl's randomized algorithm for minimum bounding circle

```

1 // Complexity: O(N) expected time
2 // Welzl's Algorithm: Smallest enclosing disk containing all points
3 // P
4
5 function circle_from_1_or_2_or_3_points(R):
6     if len(R) == 0: return Circle(origin, 0)
7     if len(R) == 1: return Circle(R[0], 0)
8     if len(R) == 2: return Circle((R[0] + R[1]) / 2, dist(R[0], R[1]) / 2)
9     return circumcircle(R[0], R[1], R[2])
10
11 function welzl(points P, boundary R, n):
12     if n == 0 or len(R) == 3:
13         return circle_from_1_or_2_or_3_points(R)
14     p = P[n - 1]
15     c = welzl(P, R, n - 1)
16     if c.contains(p): return c
17     return welzl(P, R + [p], n - 1)
18
19 function smallest_enclosing_circle(points P):
20     shuffle(P)
21     return welzl(P, [], len(P))

```

7.10 Circle Tangents

Complexity: $O(1)$

Determine up to 4 common tangent lines between 2 circles

```

1 // Complexity: O(1) time
2 // Common tangents between two circles c1(O1, r1) and c2(O2, r2)
3
4 struct Line:
5     a, b, c // ax + by + c = 0
6
7 function common_tangents(Circle c1, Circle c2):
8     d = dist(c1.O, c2.O), lines = []
9     if d < EPS and abs(c1.r - c2.r) < EPS: return INF // Infinite
10    tangents
11
12    // External tangents (r1 - r2)
13    for sign1 in [1, -1]:
14        // Angle alpha between O1->O2 and tangent
15        cos_val = (c1.r - c2.r * sign1) / d
16        if abs(cos_val) <= 1:
17            sin_val = sqrt(max(0.0, 1 - cos_val * cos_val))
18            for sign2 in [1, -1]:
19                // Construct tangent lines
20                tangent = build_tangent_line(c1, c2, cos_val,
21                    sin_val * sign2)
22                lines.append(tangent)
23    return lines

```

7.11 Geometry Formulas

Planar theorems, circles, triangle centers, circle inversion & 3D primitives

• **Linear Transformation** $\leftrightarrow$ **Matrix:** Every linear map $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ has $A = [T(e_1) \cdots T(e_n)]$ such that $T(x) = Ax$. Common 2D forms:

– Rotation by θ : $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$; scaling: $\begin{pmatrix} s_x & 0 \\ 0 & s_y \end{pmatrix}$.

– Reflect across $Ox, Oy, y = x$: $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

– Shear: $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ or $\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$; projection onto unit vector u : $P = uu^T$.

– Translation is affine, not linear: $x' = Ax + b$. In homogeneous coordinates, $\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$.

– Composition $T_2 \circ T_1$ corresponds to $A_2 A_1$; repeating one transform k times corresponds to A^k .

• **Pick's Theorem:** $S = I + \frac{B}{2} - 1 \implies I = S - \frac{B}{2} + 1$, where $B = \sum \gcd(|x_{i+1} - x_i|, |y_{i+1} - y_i|)$.

• **Euler's Planar Formula:** $V - E + F = 1 + C$ ($V - E + F = 2$ for connected planar graphs).

• **Ptolemy's Theorem:** Cyclic quad $ABCD \iff AC \cdot BD = AB \cdot CD + BC \cdot AD$.

• **Stewart's Theorem:** $b^2 m + c^2 n = a(d^2 + mn)$ (cevia d divides side a into m, n).

• **Bretschneider's Formula** (General quadrilateral area):

$$S = \sqrt{(p-a)(p-b)(p-c)(p-d) - abcd \cos^2 \left(\frac{A+C}{2} \right)}$$

(Cyclic quad $\implies \cos(\frac{A+C}{2}) = 0$, reduces to Brahmagupta's formula).

• **Descartes' Theorem** (4 mutually tangent circles):

$$\left(\sum_{i=1}^4 k_i \right)^2 = 2 \sum_{i=1}^4 k_i^2 \quad \left(k_i = \pm \frac{1}{R_i} \right)$$

• **Circle Inversion:** Center O , radius R . $P \mapsto P'$ satisfies $OP \cdot OP' = R^2$. Maps lines not through $O \longleftrightarrow$ circles through O ; preserves angles and tangencies.

• **Circle-Circle Intersection:** Let $d = |O_1 - O_2|$. Intersects when $|R_1 - R_2| \leq d \leq R_1 + R_2$, angle $\alpha = \arccos \left(\frac{R_1^2 + d^2 - R_2^2}{2R_1 d} \right)$.

• **Apollonius Circle:** Locus $\frac{PA}{PB} = k$ ($k \neq 1$) is a circle with diameter connecting the internal and external division points of AB with ratio k .

• **Circumcenter O & Orthocenter H :** Let $d_B = x_B^2 + y_B^2 - x_A^2 - y_A^2$, $d_C = x_C^2 + y_C^2 - x_A^2 - y_A^2$:

$$x_O = \frac{d_B(y_C - y_A) - d_C(y_B - y_A)}{D}$$

$$y_O = \frac{(x_B - x_A)d_C - (x_C - x_A)d_B}{D}$$

$$D = 2[(x_B - x_A)(y_C - y_A) - (x_C - x_A)(y_B - y_A)]$$

$$H = A + B + C - 2O$$

• 3D Primitives:

- 3D Cross Product: $\vec{A} \times \vec{B} = (A_y B_z - A_z B_y, A_z B_x - A_x B_z, A_x B_y - A_y B_x)$.
- Tetrahedron Volume: $V = \frac{1}{6} |((B - A) \times (C - A)) \cdot (D - A)|$.
- Plane through P_0 with normal $\vec{n}$: $\vec{n} \cdot (P - P_0) = 0$; Point-plane distance: $\frac{|\vec{n} \cdot (P - P_0)|}{|\vec{n}|}$.

7.12 AREA

Complexity: $\mathcal{O}(n \log n)$

Area of Union of Rectangles with Sweep-line and Segment Tree

```
1 // Complexity: O(N log N) time
2 // Sweep-line with Segment Tree to compute Area of Union of Axis-
  Aligned Rectangles
3
4 struct Event:
5     x, y1, y2, type (+1 for start, -1 for end)
6
7 function union_area_rectangles(rectangles):
8     events = [], Y_coords = []
9     for r in rectangles:
10         events.append(Event(r.x1, r.y1, r.y2, +1))
11         events.append(Event(r.x2, r.y1, r.y2, -1))
12         Y_coords.append(r.y1); Y_coords.append(r.y2)
13     sort(events, by x)
14     unique_sorted(Y_coords)
15
16     // Segment tree maintains covered length on Y intervals
17     segtree_init(Y_coords)
18     total_area = 0, last_x = events[0].x
19
20     for e in events:
21         total_area += segtree.total_covered_length() * (e.x -
            last_x)
22         segtree.update_range(e.y1, e.y2, e.type)
23         last_x = e.x
24     return total_area
```

7.13 Manhattan MST

Complexity: $\mathcal{O}(n \log n)$

Manhattan Minimum Spanning Tree via 8-octant sweep

```
1 // Complexity: O(N log N) time
2 // Minimum Spanning Tree under Manhattan distance: generates O(N)
  candidate edges via 8 octants
3
4 function manhattan_mst_edges(points P):
5     candidate_edges = []
6     for dir1 = 0 to 1:
7         for dir2 = 0 to 1:
8             for dir3 = 0 to 1:
9                 sort(P, by x + y)
10                // Use Fenwick tree / map to find best point
                minimizing (x_j - y_j) with y_j <= y_i
11                BIT_sweep_closest(P, candidate_edges)
12                for p in P: swap(p.x, p.y)
13                for p in P: p.x = -p.x
14                for p in P: p.y = -p.y
15    return kruskal_mst(P, candidate_edges)
```

7.14 Planar Dual Graph

Complexity: $\mathcal{O}(E \log E)$

Planar embedding to dual graph: extracts faces (including outer face) and dual edges via angular sorting on half-edges (darts).

```
1 // Complexity: O(E log E) time, O(V + E) space
2 // Constructs planar dual: extracts faces (including outer face)
  and dual edges
3
4 struct PlanarDual:
5     face_count = 0, dual_edges = []
6
7     function build(points P, edges E):
8         // 1. Build directed half-edges (darts)
9         for i, (u, v) in enumerate(E):
10             darts[u].push(2 * i) // u -> v
11             darts[v].push(2 * i + 1) // v -> u
12
13         // 2. Sort outgoing darts CCW by angle around each vertex
14         for u in 0..V-1:
15             sort(darts[u], by CCW_angle_around(u))
16             for idx, d in enumerate(darts[u]): pos[d] = idx
17
18         // 3. For each dart d = (u -> v), next dart along face
            boundary is:
19         // the dart immediately clockwise from reverse dart (v
            -> u)
20         for d in 0..2*E-1:
21             v = to[d], rev_pos = pos[d ^ 1], deg = len(darts[v])
22             next_dart[d] = darts[v][(rev_pos - 1 + deg) % deg]
23
24         // 4. Trace faces along next_dart cycles
25         for start in 0..2*E-1:
26             if face_of[start] != -1: continue
27             cur = start; face_id = face_count++
28             do:
29                 face_of[cur] = face_id
30                 cur = next_dart[cur]
31             while cur != start
32
33         // 5. Dual edge for original edge i is (face_of[2*i],
            face_of[2*i+1])
34         for i in 0..E-1:
35             dual_edges.push({face_of[2 * i], face_of[2 * i + 1]})
```

8 Misc

8.1 Template

```
1 #include <bits/stdc++.h>
2 #define FOR(i, a, b) for (int i = a; i <= b; i++)
3 #define FORD(i, a, b) for (int i = a; i >= b; i--)
4 #define endl '\n'
5 #define pb push_back
6 #define pf push_front
7 #define fastio ios_base::sync_with_stdio(NULL); cin.tie(NULL); cout
  .tie(NULL);
8
9 //Data structure
10 #define vi vector<int>
11 #define int long long
12 #define ii pair<int, int>
13 #define iii pair<int, ii>
14
15 //Bitmask (absolutely have a () outside of every thing)
16 #define LSB(x) (x & (-x))
```

```
17 #define ON(x, i) ((x >> i)&1)
18 #define OFF(x, i) !ON(x, i)
19 #define SET(x, i) (x | (1LL << i))
20 #define UNSET(x, i) (SET(x, i) ^ (1LL << i))
21
22 #define TASKNAME "test"
23
24 template<typename T> bool maximize(T &a, T b){ if (a < b){ a = b
  ; return true; } return false; }
25 template<typename T> bool minimize(T &a, T b){ if (a > b){ a = b
  ; return true; } return false; }
26 using namespace std;
27
28 const int MOD = 1e9 + 7;
29 void add(int &a, int b){
30     a += b;
31     if (a >= MOD) a -= MOD;
32 }
33
34 void sub(int &a, int b){
35     a -= b;
36     if (a < 0) a += MOD;
37 }
38 void solve(){
39
40 }
41 main(){
42     fastio;
43     if (fopen(TASKNAME".inp", "r")){
44         freopen(TASKNAME".inp", "r", stdin);
45         freopen(TASKNAME".out", "w", stdout);
46     }
47     int t = 1;
48     //cin >> t;
49     while(t--){
50         solve();
51     }
52 }
```

8.2 Debugging Magic

```
1 #define what_is(x) cerr << #x << " = " << x << endl;
2 #define deb(...) logger(#__VA_ARGS__, __VA_ARGS__)
3 template<typename ...Args>
4 void logger(string vars, Args&&... values) {
5     cout << "{" << vars << " } = {";
6     string delim = "";
7     (... , (cout << delim << values << "}", delim = ", "));
8 }
```

8.3 Expression Parsing

```
1 // Complexity: O(N) time and space
2 // Shunting-Yard algorithm: Parses infix arithmetic expressions
  with operator precedence
3
4 function parse_expression(string expr):
5     stack<double> values; stack<char> ops
6
7     function apply_op():
8         op = ops.pop(); b = values.pop(); a = values.pop()
9         values.push(eval_op(a, b, op))
10
11     for token in tokenize(expr):
12         if is_number(token):
13             values.push(token)
14         else if token == '(':
15             ops.push('(')
```

```

16     else if token == ')':
17         while ops.top() != '(': apply_op()
18         ops.pop() // remove '('
19     else: // Operator +, -, *, /, ^
20         while not ops.empty() and precedence(ops.top()) >=
precedence(token):
21             apply_op()
22         ops.push(token)
23
24 while not ops.empty(): apply_op()
25 return values.top()

```

8.4 Johnson Scheduling

Scheduling on two machines

```

1 // Complexity: O(N log N) time
2 // Johnson's Rule: Optimal scheduling of N jobs on 2 machines in
sequence (Machine 1 then Machine 2)
3
4 struct Job:
5     id, a (time on M1), b (time on M2)
6
7 function johnson_scheduling(jobs):
8     // Partition jobs into Group 1 (a_i <= b_i) and Group 2 (a_i >
b_i)
9     G1 = [j for j in jobs if j.a <= j.b]
10    G2 = [j for j in jobs if j.a > j.b]
11
12    sort(G1, ascending by a)
13    sort(G2, descending by b)
14
15    return G1 + G2 // Optimal processing order

```

8.5 Monte Carlo & Las Vegas Framework

- **Monte Carlo:** Fixed runtime, bounded error $\leq \epsilon$. Amplified by k independent trials $\implies$ error drops exponentially $\leq (1-p)^k$ (e.g. 2^{-k}).
- **Las Vegas:** Guaranteed 100% correct, randomized runtime. Expected iterations = $1/p$ with verification cost $O(T_{\text{check}})$.
- **Freivalds' Verification** ($AB \stackrel{?}{=} C$): Compute $P = A(Bv) - Cv$ for random $v \in \{0,1\}^N$ in $O(N^2)$. If $AB \neq C$, $\Pr(P = 0) \leq 1/2$. Running k trials $\implies$ error $\leq 2^{-k}$.
- **Schwartz-Zippel Lemma (PIT):** For non-zero poly $P(x_1, \dots, x_n)$ of degree d , evaluating at random $r_i \in S$ gives $\Pr(P(r) = 0) \leq d/|S|$. Basis of Tutte matching & polynomial identity testing.

8.6 Calendar

```

1 // Complexity: O(1) time
2 // Zeller's Congruence: Computes day of the week (0 = Sunday, 1 =
Monday, ..., 6 = Saturday)
3
4 function day_of_week(year, month, day):
5     if month < 3:
6         month += 12; year -= 1
7     K = year % 100, J = year / 100
8     // Zeller's formula for Gregorian calendar
9     h = (day + (13 * (month + 1)) / 5 + K + K / 4 + J / 4 + 5 * J)
% 7
10    return (h + 6) % 7 // Shift so 0 = Sunday

```

8.7 Josephus

```

1 // Complexity: O(N) for linear, O(K log N) for large N, small K
2 // Josephus Problem: 0-indexed survivor among N people eliminating
every K-th person
3
4 function josephus_linear(N, K):
5     ans = 0
6     for i = 2 to N:
7         ans = (ans + K) % i
8     return ans
9
10 function josephus_fast(N, K):
11     if N == 1: return 0
12     if K == 1: return N - 1
13     if K > N: return (josephus_fast(N - 1, K) + K) % N
14     cnt = N / K
15     res = josephus_fast(N - cnt, K)
16     res -= N % K
17     if res < 0: res += N
18     else: res += res / (K - 1)
19     return res

```

8.8 Interactive Tricks

```

1 // QUERY WRAPPER
2 ask(args):
3     print("?", args); flush()
4     reply = read(); queries++
5     if reply is invalid or -1: exit()
6     return reply
7
8 // BINARY SEARCH: invariant ask(lo)=false, ask(hi)=true
9 while lo + 1 < hi:
10    mid = (lo + hi) / 2
11    if ask(mid): hi = mid
12    else: lo = mid
13
14 // THREE OUTCOMES -> ceil(log_3 N)
15 while |candidate| > 1:
16    split candidate evenly into A, B, C
17    res = compare(A, B)
18    candidate = (res favors A ? A : res favors B ? B : C)
19 // Valid only if one target and equal known baselines for A, B.
20
21 // DISCRETE INTERMEDIATE VALUE (array need not be monotone)
22 // Given |a[i]-a[i-1]|=1 and a[L] <= x <= a[R]:
23 mid = (L + R) / 2
24 if ask(mid) < x: L = mid // x must occur on the right
25 if ask(mid) > x: R = mid // x must occur on the left
26
27 // TOURNAMENT: MAXIMUM + SECOND MAXIMUM
28 for each match (u, v):
29     winner beats loser; defeated[winner].push(loser)
30 maximum = tournament winner
31 secondMaximum = max(defeated[maximum])
32 // n-1 + ceil(log2 n)-1 comparisons with a balanced bracket.
33
34 // PACK MANY TESTS INTO ONE QUERY
35 Y = product(selected prime powers)
36 g = gcd_query(X, Y) // reveals all selected factors in X
37 // General recipe: recover counts/factors first, then order/
exponents.
38 // If total is known, query/build cheap categories and infer the
last one.
39
40 // CANONICALIZE UNKNOWN STATE
41 // Prefer forcing every possible x to known y over identifying x.
42 mul(10^d - 1); digit() // S(x(10^d-1)) = 9d, 1 <= x <= 10^d
43 add(target - 9d)
44

```

```

45 // COST-AWARE MAXIMUM: charge counts only when true max is queried
46 (best, second) = first two items in decreasing order
47 for u in random_permutation(remaining):
48     if compare(u, second) == u:
49         if compare(u, best) == u: second = best, best = u
50         else: second = u
51
52 // PEAK SEARCH / QUERY REUSE
53 cache every ask(position)
54 maintain a witness that some peak remains in [L, R]
55 query reusable interior points (golden section) or a separator row/
column
56 move toward a larger neighbor; large budget: random starts -> best
-> hill climb
57
58 // LOCAL CONSTRAINT PROPAGATION (Minesweeper-like)
59 repeat until no change:
60     rem = clue - knownMines; unk = unknownNeighbors
61     if rem == 0: mark all unk safe
62     if rem == |unk|: mark all unk mines
63 // Then subtract overlapping equations or enumerate only a small
frontier.
64
65 // ALGEBRA / RANDOMIZATION
66 x + y = (x AND y) + (x OR y)
67 x XOR y = (x OR y) - (x AND y)
68 Pr[miss bad set after t samples] <= (1-k/n)^t
69
70 // FINAL: legal/distinct indices, query budget, one flushed final
answer.

```

8.9 Random, Bit Tricks, Bitset

```

1 // Complexity: O(1) or O(3^N) for submask iteration
2 // Useful Bit Hacks, Bitset, and Submask Iterations
3
4 // 1. Iterate all submasks of a given mask:
5 for sub = mask; sub > 0; sub = (sub - 1) & mask:
6     process(sub)
7 process(0)
8
9 // 2. Gosper's Hack: Iterate all subsets of size K from N elements
10 function gosper_hack(K, N):
11     mask = (1 << K) - 1
12     while mask < (1 << N):
13         process(mask)
14         c = mask & -mask, r = mask + c
15         mask = (((r ^ mask) >> 2) / c) | r
16
17 // 3. Bit Tricks:
18 // Lowest set bit: x & -x
19 // Clear lowest set bit: x & (x - 1)
20 // Builtin fast operations:
21 // __builtin_clz(x) (leading zeros), __builtin_ctz(x) (trailing
zeros), __builtin_popcountll(x)
22
23 // 4. Bitmask DP State Reduction (Eliminate step/size loop):
24 // In matching/permutations: step i is implicitly
__builtin_popcount(mask).
25 // dp[i][mask] -> dp[mask], reducing space from O(N * 2^N) to O(2^N
).
26 // Iterating mask = 0..(1<<N)-1 guarantees topological DAG order
without outer loop for i.

```

8.10 Transformations

- **When Stuck:** Re-read statement! Formulate small cases ($N \leq 6$) to find patterns/invariants; deduce complexity ($N \leq 20 \rightarrow 2^N, 500 \rightarrow N^3, 5000 \rightarrow N^2, 2 \cdot 10^5 \rightarrow N \log N$).

- **Opt** → **Decision (Binary Search)**: If predicate $P(X)$ is monotonic, binary search over X to solve feasibility.
- **Global Permutation** → **Local Swap**: Greedy exchange argument: swapping $(i, j) \rightarrow (j, i)$ improves objective $\iff f(i, j) < f(j, i)$.
- **Constraints** → **Graph / Flow / Cut**: 2-SAT, difference constraints ($x_j - x_i \leq w$), Project Selection (Min-Cut), Hall's Marriage, Dilworth (DAG path cover = $|V| - \text{MaxMatching}$).
- **Algebraic** → **Geometry / Convexity**: Upper/lower envelopes, Convex Hull, Slope Trick, Minkowski sum, CHT.
- **Weighted Average** → **Convex Hull**: Think of λ_i as mixing ratios ($\lambda_i \geq 0, \sum_i \lambda_i = 1$). Then $x = \sum_i \lambda_i x_i$ stays between the inputs: two points give segment AB , three points give the filled triangle, and many points give their convex hull. Conversely, every point in the convex hull has such a representation. In 1D, $x \in [\min x_i, \max x_i]$. Normalize positive weights by $\lambda_i = w_i / \sum_j w_j$.
- **Ratio of (Weighted) Totals** → **Average of Ratios**: For $w_i \geq 0, b_i > 0, \frac{\sum_i w_i a_i}{\sum_i w_i b_i}$ is a weighted average of a_i/b_i with new weight $w_i b_i$. Hence it lies in $[\min_i (a_i/b_i), \max_i (a_i/b_i)]$. The unweighted case is $w_i = 1$.
- **Mix First** → **Jensen**: For convex f (bowl-shaped), evaluating after mixing is cheaper: $f(\sum_i \lambda_i x_i) \leq \sum_i \lambda_i f(x_i)$. Read it as $f(\text{average}) \leq \text{average of } f$.
- **Global Range** → **Divide & Conquer**: Split $[l, r] \rightarrow [l, m]$ and $[m+1, r]$ + cross contributions (CDQ D&C, D&C DP).
- **Offline Deletions** → **Additions (Time Reversal)**: Invert query timeline $Q \rightarrow 1$ to turn deletions into DSU additions. Game DP: work backward from terminal state.
- **Search Space Split (Meet-in-Middle)**: Split $N = N/2 + N/2$, reducing $2^N \rightarrow 2^{N/2}$ via hash table / two pointers.
- **Exact** ($=k$) → **Lagrange Penalty** ($\geq k$): Alien's Trick (WQS) relaxes exact k on convex DP; Binomial Inversion / PIE converts exact k to $\geq k$.
- **Total Evaluation** → **Element Contribution**: Invert perspective: sum contribution of each element/edge/pair (u, v) over all valid states (Fubini).
- **Threshold Partitioning** ($\sqrt{N}$ Heavy/Light): Elements $\leq B$ updated statically/cached; elements $> B$ (at most N/B) processed dynamically (Mo's algorithm).
- **State Compression**: In DP, step count $i = \text{popcount}(\text{mask}) \implies dp[i][\text{mask}] \rightarrow dp[\text{mask}]$, preserving topological order.

- **Exact Count** → **Complementary / PIE**: Count valid states via Total – Invalid, Stars & Bars, PIE.
- **Impartial Game** → **Nim Sum**: Transform games on DAG into XOR sum of Sprague-Grundy values ($SG(u) = \text{mex}\{SG(v)\}$).
- **Undirected Vertex Geography** → **Matching**: A token starts at s ; players alternately move it to an unvisited neighbor, and the vertex just left is deleted. The first player wins iff s belongs to every maximum matching of G , equivalently $\nu(G) > \nu(G - s)$.
- **General Case** → **Simplified Special Case**: Solve easier restrictions first ($A_i \in \{0, 1\}$, tree, $N \leq 6$) to reveal pattern before generalizing.
- **Exhaustive Search** → **Branch & Bound**: Prune DFS subtrees with admissible bounding heuristics.
- **Manhattan** ↔ **Chebyshev** (45° Rotation): $(x, y) \mapsto (x+y, x-y) \implies |x_1 - x_2| + |y_1 - y_2| = \max(|x'_1 - x'_2|, |y'_1 - y'_2|)$. Decouples 2D dimensions into independent 1D axes!
- **Max/Min & Absolute Value Decomp**: $\max(A, B) = \frac{A+B+|A-B|}{2}$, $\min(A, B) = \frac{A+B-|A-B|}{2}$; $|x-a| = \max(x-a, a-x)$. To minimize $\sum |x-a_i| + |y-b_i|$, split by 4 sign pairs $(\pm x \pm y)$.
- **Threshold / Layer Counting (Fubini)**: $\sum_{v=1}^{\max A} A_i = \sum_{v=1}^{\max A} \text{count}(A_i \geq v)$ (Summing values $\longleftrightarrow$ Counting elements $\geq v$).
- **Square** → **Sum of Odd Numbers**: $x^2 = 1+3+5+\dots+(2x-1) = \sum_{i=1}^x (2i-1)$.
- **Repeated Ones** → **Geometric Sum**: $\underbrace{11\dots1}_{i \text{ digits}} = 1 + 10 + \dots + 10^{i-1} = \frac{10^i-1}{9}$.
- **Increasing / Decreasing Sequence** → **Nonnegative Variables**: To remove coupled order constraints, encode adjacent gaps. If $A_1 \leq \dots \leq A_n$, set $D_i = A_i - A_{i-1} \geq 0$; if $A_1 \geq \dots \geq A_n$, set $D_i = A_{i-1} - A_i \geq 0$ ($i \geq 2$). Strict order means $D_i \geq 1$. Recover by $A_i = A_1 + \sum_{j=2}^i D_j$ (increasing) or $A_i = A_1 - \sum_{j=2}^i D_j$ (decreasing).
- **Range Add** → **Difference Array / Prefix Sums**: $A_i = \sum_{j=1}^i D_j$ ($D_i = A_i - A_{i-1}$); Range add $[l, r] \rightarrow 2$ point updates on D . Static sums $\rightarrow S_r - S_{l-1}$.
- **Range / All-Pairs** → **Adjacent Condition**: $\min_{i \neq j} |A_i - A_j| = \min_i (A_{i+1} - A_i)$ after sort; $A_j - A_i \geq (j-i)D \iff A'_{i+1} \geq A'_i$ ($A'_i = A_i - iD$); Majority ($> 50\%$) subsegment ≥ 2 always contains length 2 (x, x) or 3 (x, y, x) witness.

- **Product to Sum**: $ab = \binom{a+b}{2} - \binom{a}{2} - \binom{b}{2}$; $\prod A_i \rightarrow \sum \log A_i$ (or log-probabilities).
- **Bitwise Identities**: $A+B = (A \oplus B) + 2(A \text{ AND } B) = (A \text{ OR } B) + (A \text{ AND } B)$; $A \oplus B = (A \text{ OR } B) - (A \text{ AND } B)$; Adjacent bit flips: $\sum_{i=1}^n \text{popcount}((i-1) \oplus i) = 2n - \text{popcount}(n)$.
- **Random Witness Search (Monte Carlo** ↔ **Las Vegas)**: When valid solutions occupy density $\geq p$: (1) **Monte Carlo**: k random samples miss all with prob $\leq (1-p)^k$ ($p=1/2, k=40 \implies \text{error} < 10^{-12}$). (2) **Las Vegas**: If verifiable in $O(T)$, sample and verify $\implies$ expected $1/p$ trials, 100% correct in expected $O(T/p)$ time (e.g. $> n/2$ majority in expected $O(n)$, quickselect, random pivot).
- **Set / Multiset** → **Monte Carlo Fingerprinting**: Map $v \mapsto r_v \in [0, 2^{64}-1]$. Sets via $\bigoplus r_v$, multisets via $\sum \text{cnt}(v)r_v \pmod{2^{64}}$ or polynomial hash (Schwartz-Zippel). Collision prob $\leq 2^{-64}$.
- **Algebraic Existence** → **Schwartz-Zippel (Monte Carlo)**: Matching (Tutte Matrix), spanning trees (Kirchhoff). Replace symbolic edges by random $r \in \mathbb{F}_p$ ($p \approx 10^9 + 7$); compute det in $O(V^3)$. $\Pr(\text{false zero}) \leq V/p$.
- **Color Coding** ($k \leq 5$) [**Monte Carlo**]: Randomly color N vertices with k colors. Prob target k -subgraph is colorful is $k!/k^k \geq e^{-k}$. Repeat $O(e^k \ln(1/\epsilon)) \approx 300$ trials for error $< \epsilon$.
- **Dirichlet / Pigeonhole (4-Sum** $a_x + a_y = a_z + a_w$): When $N > \sqrt{2} \max A$, 2 pairs with equal sums always exist since $\binom{N}{2} > 2 \max A$. Find in $O(N^2)$ by hashing pairs.
- **Boyer-Moore Majority Vote**: Find majority element ($> n/2$ occurrences) in $O(n)$ time and $O(1)$ space with a single counter. Sperner's antichain max size: $\binom{n}{\lfloor n/2 \rfloor}$.

8.11 Checked

- **Read constraints and edge cases** ($N=0, 1$, negative, disconnected)?
- Multiple test cases reset / clear global state properly?
- Array / 1-based vs 0-based indexing bounds?
- Integer overflow (use `long long / __int128`)?
- Functions missing `return` statements (Undefined Behavior)?
- Floating point precision / epsilon comparisons (`eps = 1e-9`)?
- Fast I/O included (`cin.tie(NULL), 'n'` instead of `endl`)?